

Higher-order multi-scale randomized neural network (HOMS-RNN) method for high-accuracy and efficient heat conduction simulation of composite materials

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Abstract

The prohibitive computation and Frequency Principle dominate significant simulation challenges for deep learning methods to finely resolve multi-scale problems. In this study, an innovative higher-order multi-scale randomized neural network (HOMS-RNN) framework is proposed to accurately and efficiently compute transient thermal problems of composite materials with highly discontinuous and high-contrast material parameters, which inherits the respective high-accuracy, high-efficiency, mesh-free and space-time advantages of higher-order multi-scale approach and randomized neural network. To address the large complexity and Frequency Principle of multi-scale problems, higher-order multi-scale approach is first employed to decouple and reduce complicated multi-scale thermal problem of heterogeneous materials as a simpler macroscopic homogenized problem and some small-scale lower-order and higher-order microscopic cell problems. Then, by utilizing macroscopic homogenized problem and microscopic cell problems as new physical constraints of neural networks, randomized neural networks are designed to efficiently solve microscopic unit cell functions and macroscopic homogenization equations for further assembling into higher-order multi-scale asymptotic solutions of multi-scale heat problems by employing automatic differentiation approach. Moreover, the error estimate of the proposed HOMS-RNN method is rigorously demonstrated under some hypotheses. Finally, the favorable accuracy and efficiency of the proposed HOMS-RNN method are validated through a series of multi-scale thermal simulations compared with finite element method and multi-scale method, including high-contrast composites, three-dimensional composites and porous materials.

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This study also offers an efficient HOMS-RNN computational framework that has strong potential and is competitive with traditional numerical methods for large-scale heat simulations of complicated composite materials.

Keywords: Multi-scale heat equations, Higher-order multi-scale constraints, Efficient randomized neural network, Error estimate, Highly heterogeneous materials

1. Introduction

Natural and artificial composite materials are extensively used in aerospace, aviation and a range of other applications owing to their excellent thermal properties. The composite materials generally possess highly heterogeneous microstructure and significant property difference of their component materials, which result in complicated multi-scale behaviors and responses of composite materials. Especially in the fields of aerospace and aviation, accurate and efficient simulation of thermal transfer behaviors of composite materials is vital to design and optimize engineering structures manufactured by composite material [1]. However, traditional numerical approaches will require a heavy computational burden for full fine-scale computation of composite materials. As for some large-scale engineering structures, traditional numerical approaches cannot guarantee computational convergence and even exceed the computing power of modern computers.

In the last few decades, scientists and engineers have developed a variety of multi-scale numerical approaches to accurately and efficiently simulate multi-scale phenomena in heterogeneous materials, including asymptotic homogenization method (AHM) [2], heterogeneous multi-scale method (HMM) [3, 4], variational multi-scale method (VMS) [5], multi-scale finite element method (Ms-FEM) [6], generalized multi-scale finite element methods (GMsFEM) [7, 8] and localized orthogonal decomposition method (LOD) [9], etc. Among these multi-scale methods, AHM is the pioneer approach for solving multi-scale problems. By the asymptotic homogenization process, a complicated multi-scale system of composite material is approximated by a simple homogenized equation defined on an equivalent homogeneous material at macro-scale and some small-scale microscopic cell equation defined on an unit cell (UC) or representative volume element (RVE) at micro-scale, and then can be numerically solved in a fast manner. Moreover, AHM has been extensively utilized in engineering applications due to its rigorous mathematical theory, definite physical significance and also can be well integrated with classical finite element method. Nevertheless, through extensive numerical calculations and engineering practices, Cui et al. [10, 11, 12, 13, 14] found that classical AHM can't precisely capture the highly oscillatory response of inhomogeneous materials. In order to accurately simulating and analyzing the physical behaviors and responses of composite materials, Cui et al. [13, 14, 15, 16, 17] developed a novel higher-order multi-scale method by devising new higher-order microscopic cell functions and corresponding correction terms to keep the balance of locally physical quantity we are concerned

about, that can substantially enhance the computing accuracy than conventional AHM. Nevertheless, it is worth emphasizing that all above-mentioned numerical methods demand time-consuming mesh discretization and the computational accuracy of the numerical results quite relies on the mesh quality, especially for complex regions, large deformations or moving interfaces. Other than mesh quality, these numerical approaches fall into the curse of dimensionality when simulating high-dimensional multi-scale problems. Motivated by theoretical research and engineering application, scientists aims to develop new computational methods based on machine learning and deep learning techniques for overcoming the shortcomings of conventional multi-scale methods.

With the rapid development of computer technology, leveraging the deep learning methods to assist scientific computing recently has been an emerging trend. Various works have been employed neural networks to approximate and solve the solutions of partial differential equations (PDEs), such as physics-informed neural network (PINN) [18], deep Ritz method (DRM) [19], deep Galerkin method (DGM) [20], deep energy method (DEM) [21] and a variety of their revised versions [22, 23, 24, 25, 26, 27, 28, 29, 30], etc. The remarkable capabilities and mesh-free advantage of these methods have been demonstrated for solving high-dimensional PDEs and handling problems with complex geometry, which make these neural network-based approaches become powerful alternatives to conventional numerical methods and widely used in various engineering contexts. However, one biggest dilemma of conventional neural network-based approaches is that training processes to determining neural networks are time-consuming since they involve solving nonlinear optimization problems, which are computationally expensive and may get stuck in local optima [31]. To overcome the challenges in terms of training efficiency and accuracy, exploiting randomization in neural networks has a long history, whose core idea is that a subset of the network parameters is assigned to random values and fixed (not trainable) while the rest of the network parameters are trained [32, 33, 34, 35, 36, 37, 38, 39]. Randomized neural network (RNN) or extreme learning machine (ELM) is a special type of deep neural networks, which do not need gradient descent-type optimization solvers during training process. Specifically, RNN works by randomly selecting and fixing all biases and weights except for those in the last layer, which results in a linear combination. Other weights of the last layer for determining RNN are then computed by solving a least-square problem. On the basis of original RNN, some new approaches have been developed recently for solving different kinds of PDEs. In reference [34], Dong and Li developed a neural network-based approach, which combines ELMs and domain decomposition method to solve linear and nonlinear PDEs. Furthermore, Wang and Dong extended ELM approach from low to high dimensions [35]. Numerical experiments indicated that the proposed ELM method is both cost-effective and more accurate for high-dimensional PDEs compared with the PINN method. In addition, Wang et al. incorporated RNNs with weak formulations, utilizing the Petrov-Galerkin method [36, 37] and discontinuous Galerkin method [38, 39] respectively, to solve PDEs. These RNN-based methods can achieve higher accuracy with fewer degrees of freedom for solving PDEs, and can also

solve time-dependent problems in the space-time approach precisely and efficiently. In a summary, leveraging the deep learning methods to assist scientific computing is of great theoretical significance and practical value.

Despite various neural network-based approaches have been established for numerically solving mathematical and engineering problems related to PDEs, only a few works about neural network-based approaches focus on solving multi-scale problems. From the existing researches in [40, 41], the Frequency Principle to solve multi-scale problems has been revealed, that is, deep neural networks often fit objective functions from low to high frequencies and exhibit slower convergence for higher frequencies for various scientific computing problems. The Frequency Principle results in one significant challenge for accurately and efficiently computing multi-scale problems. Additionally, to accurately capture the locally oscillatory information of the solutions of multi-scale PDEs, a large amount of randomly residual points are required in computational domain, which leads to a large number of neurons in the input layer of neural networks. Hence, the training process of corresponding neural network based on optimization solvers is very slow or even not converge [42, 43]. To address the above-mentioned challenges for solving multi-scale problems by deep learning methods, researchers have developed some novel neural network-based approaches motivated by enormous engineering values in the fields of material science, solid mechanics, fluid mechanics and chemical engineering, etc. In references [42, 43], Xu and Zhang et al. established the multi-scale deep neural networks (MscaledNNs) to simulate multi-scale elliptic PDEs, and further combined subspace decomposition techniques to develop new SD²NN method for solving elliptic type multi-scale PDEs. In reference [44], Efendiev and Wang et al. designed novel multi-layer neural networks, which employ deep learning techniques combined with local multi-scale model reduction methodologies to predict flow dynamics. In addition, Leung and Lin et al. presented a neural homogenization based PINN (NH-PINN) in [45], which can effectively improve the numerical accuracy of PINN for solving multi-scale problems assisted with homogenization technique. Recently, a new explicit-implicit splitting learning approach was proposed in [46], which provides computational saving without sacrificing accuracy for multi-scale problems with time-dependent source. In reference [47], an adaptive multi-scale neural network with Resnet blocks (adaptive-MS-Resnet) architecture was developed for computing the Poisson equation, Helmholtz equation, wave equation and Klein-Gordon equation. In this adaptive-MS-Resnet, the radial scaling was employed to capture the oscillatory part of the PDEs' solution. Furthermore, Han and Lee proposed a neural network-based method to implement numerical homogenization of multi-scale problems [48], whose outstanding advantage is free to design the hand-crafted neural network architecture and the cell problem for calculating the homogenization coefficient. Combining periodic homogenization theory and physics-informed neural networks, Chen et al. [49] proposed a physically informed deep homogenization neural network to compute the homogenized properties, local temperature and gradient distributions of composite materials. Nevertheless, this work only implemented the computation of low-contrast composites with a thermal conductivity ratio of 4:1. Furthermore, Chen et al.

extended the proposed deep homogenization neural network to predict the local stress field and homogenized moduli of heterogeneous materials with two- and three-dimensional periodicity [50]. In reference [51], authors proposed a neural network approach assisted fast multi-scale analysis for the elastic properties prediction of 3D braided composites under uncertainty caused by the automated manufacturing procedure. In reference [52], a machine learning approach based on convolutional neural networks (CNNs), was developed as a computationally efficient solution method for predicting the homogenized macroscopic stress. In reference [53], feedforward neural network (FNN) and recurrent neural network were employed to model the multi-scale constitution and effectively predict the materials behavior along unknown paths. In order to effectively predict the effective material properties of real composite materials, Dong et al. [54, 55] proposed a novel wavelet-neural network framework that introduced wavelet decomposition into deep neural networks (DNNs) to efficiently and precisely evaluate the mechanical and thermal parameters of the composites. This method only involve computing equivalent material parameters. Furthermore, Dong et al. [56] developed a higher-order multi-scale deep Ritz method (HOMS-DRM) to mesh-free solve thermal conduction problems of heterogeneous materials, which can surmount restrictions of prohibitive computation and Frequency Principle when deep learning simulations are directly employed. Although this HOMS-DRM can precisely catch the highly oscillatory information of inhomogeneous materials at micro-scale, the computing efficiency of the proposed HOMS-DRM still needs to be improved since it can not span from the limitations of gradient descent-type optimization solvers. Motivated by great engineering value of composite materials, the development of efficient and high-accuracy NN-based approach for multi-scale problems of heterogeneous materials with highly discontinuous and high-contrast material parameters has become a research hotspot and has some urgent issues that need to be addressed.

In this work, we propose a higher-order multi-scale randomized neural network (HOMS-RNN) method, to address the Frequency Principle, prohibitive computation, the inaccuracy for capturing the microscopic oscillation and the inefficiency of optimization solvers in multi-scale heat simulation of composite materials. The proposed HOMS-RNN approach inherits the respective high-accuracy, high efficiency and mesh-free superiorities of higher-order multi-scale approach and randomized neural network method. Firstly, the low-accuracy, inefficiency and prohibitive computation of NN-based approaches for multi-scale problems lie in that multi-scale PDEs are directly used as physical constraints of NN-based approaches. Meanwhile, a large amount of randomly residual points are required in computational domain, which results in Frequency Principle. To address this challenge, higher-order multi-scale method is employed to decouple and reduce complicated multi-scale thermal problem of heterogeneous materials as a simpler macroscopic homogenized problem and some small-scale lower-order and higher-order microscopic cell problems, which are employed as new physical constraints for NN-based approaches. Moreover, another noteworthy strength of higher-order multi-scale constraints is that it can accurately capture highly oscillatory information in the microscopic scale of heterogeneous materials.

Furthermore, RNN method without gradient descent-type optimization solver is applied to efficiently and mesh-free solve macroscopic homogenized problem and microscopic cell problems, particularly in space-time solving time-dependent macroscopic homogenized problem. In addition, hard-constraint boundary condition is applied to precisely solve elliptic type microscopic cell problems with highly discontinuous and high-contrast material parameters. The main contributions of this paper are summarized as follows.

- Employing higher-order multi-scale approach to decouple original constraints (multi-scale PDEs) of neural networks as new constraints (macroscopic homogenized PDEs and microscopic cell PDEs) of neural networks, which resolve the prohibitive computation and Frequency Principle, and possess the high-accuracy computational performance for multi-scale thermal simulation.
- Developing a mesh-free and efficient RNN method to solve macroscopic homogenized PDEs and microscopic cell PDEs, particularly in space-time solving time-dependent macroscopic homogenized PDEs.
- Introducing hard-constraint boundary condition to assist HOMS-RNN method, that substantially improve the numerical accuracy for solving elliptic type microscopic cell problems and parabolic type macroscopic homogenized problem of high-contrast composite materials.
- Demonstrating the theoretical convergence of the HOMS-RNN method for multi-scale heat problems.

The remainder of the paper is structured as follows. In Section 2, we employ the higher-order multi-scale method to establish new physical constraints of neural networks for heat conduction problems of heterogeneous materials. Next, the proposed HOMS-RNN framework and corresponding algorithmic procedures will be presented at length in Section 3. After that, Section 4 gives the rigorous proof of the convergence for the presented HOMS-RNN approach under appropriate hypotheses. Furthermore, several numerical experiments including 2D composites with high-contrast material parameters, 3D composites and porous materials are conducted to verify the accuracy, efficiency and robustness of the proposed HOMS-RNN approach in Section 5. Conclusions and outlooks are presented in Section 6.

For narrative brevity, Einstein summation convention is utilized for repetitive indices throughout this study.

2. Higher-order multi-scale model for composite materials

2.1. Transient multi-scale heat conduction problem

This study considers the following parabolic equation with periodically oscillatory and highly discontinuous coefficients for describing the heat conduction

behavior of composite materials.

$$\begin{cases} \rho^\epsilon(\mathbf{x})c^\epsilon(\mathbf{x})\frac{\partial T^\epsilon(\mathbf{x},t)}{\partial t} - \frac{\partial}{\partial x_i}\left(k_{ij}^\epsilon(\mathbf{x})\frac{\partial T^\epsilon(\mathbf{x},t)}{\partial x_j}\right) = f(\mathbf{x},t), & \text{in } \Omega \times (0, T^*), \\ T^\epsilon(\mathbf{x},t) = \hat{T}(\mathbf{x},t), & \text{on } \partial\Omega \times (0, T^*), \\ T^\epsilon(\mathbf{x},0) = \check{T}(\mathbf{x}), & \text{in } \Omega, \end{cases} \quad (1)$$

where $T^\epsilon(\mathbf{x},t) \in \mathbb{R}$ is the unresolved temperature solution, $T^* > 0$ is the final time, and Ω is a bounded convex domain in $\mathbb{R}^N$ ($N = 2, 3$) with a outer boundary $\partial\Omega$. ϵ is the characteristic size of microscopic unit cell. Moreover, $\rho^\epsilon(\mathbf{x})$ is the mass density, $c^\epsilon(\mathbf{x})$ is the specific heat, $\{k_{ij}^\epsilon(\mathbf{x})\}$ is a second-order tensor that represents the thermal conductivity coefficient of the material, $f(\mathbf{x},t)$ denotes the internal heat source. In addition, it is worth noting that the superscript ϵ in multi-scale PDE (1) indicates the spatial size of microscopic unit cell (UC) of composite materials, which result in highly oscillatory and discontinuous material parameters. The detailed schematic of composite material is illustrated in Fig. 1. This multi-scale feature arises the significant challenges in the numerical simulation for composite materials.

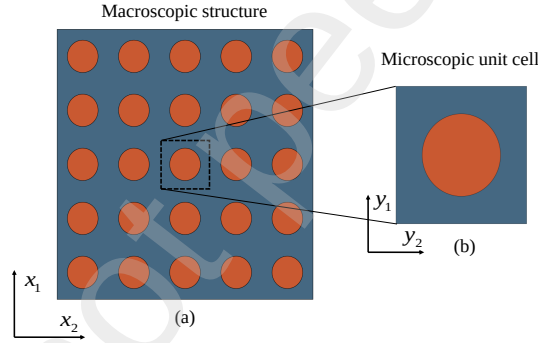


Figure 1: The schematic of composite material. (a) The macroscopic structure Ω . (b) The microscopic unit cell Y .

To ensure the well-posed property and multi-scale asymptotic analysis of transient heat conduction equation (1) of composite materials, we make the following assumptions for multi-scale PDE (1) [57].

(A₁) $k_{ij}^\epsilon(\mathbf{x})$ is symmetric, and there exist two positive constant γ_0 and γ_1 independent of ϵ such that

$$k_{ij}^\epsilon(\mathbf{x}) = k_{ji}^\epsilon(\mathbf{x}), \gamma_0|\boldsymbol{\xi}|^2 \leq k_{ij}^\epsilon(\mathbf{x})\xi_i\xi_j \leq \gamma_1|\boldsymbol{\xi}|^2,$$

where $\boldsymbol{\xi} = (\xi_1, \dots, \xi_N)$ represents an arbitrary vector with real elements in $\mathbb{R}^N$.

(A₂) $\rho^\epsilon(\mathbf{x})$, $c^\epsilon(\mathbf{x})$ and $k_{ij}^\epsilon(\mathbf{x})$ represent scalar functions belonging to $L^\infty(\Omega)$, and there are constants ρ^0 , ρ^1 , c^0 and c^1 such that

$$0 < \rho^0 \leq \rho^\epsilon(\mathbf{x}) \leq \rho^1, \quad 0 < c^0 \leq c^\epsilon(\mathbf{x}) \leq c^1.$$

- (A₃) Let microscopic variable $\mathbf{y} = \mathbf{x}/\epsilon$, and functions $\rho^\epsilon(\mathbf{x})$, $c^\epsilon(\mathbf{x})$ and $k_{ij}^\epsilon(\mathbf{x})$ are 1-periodic with regard to microscopic variable $\mathbf{y}$.
(A₄) $f(\mathbf{x}, t) \in L^2(\Omega \times (0, T^*))$.

2.2. Higher-order multi-scale model for multi-scale parabolic PDE

To resolve the Frequency Principle and prohibitive computation involving direct deep learning simulation, higher-order multi-scale model is established for time-dependent heat conduction simulation of composite materials, which is employed as new physical constraints for deep learning simulation.

According to the between macroscopic and microscopic coordinates, the following chain rule holds

$$\frac{\partial \Psi^\epsilon(\mathbf{x}, t)}{\partial x_i} = \frac{\partial \Psi(\mathbf{x}, \mathbf{y}, t)}{\partial x_i} + \frac{1}{\epsilon} \frac{\partial \Psi(\mathbf{x}, \mathbf{y}, t)}{\partial y_i}. \quad (2)$$

Afterwards, we seek for the temperature solution $T^\epsilon(\mathbf{x}, t)$ of multi-scale transient PDE (1) in the following asymptotic expansion expression

$$T^\epsilon(\mathbf{x}, t) = T^{(0)}(\mathbf{x}, \mathbf{y}, t) + \epsilon T^{(1)}(\mathbf{x}, \mathbf{y}, t) + \epsilon^2 T^{(2)}(\mathbf{x}, \mathbf{y}, t) + O(\epsilon^3). \quad (3)$$

Next, we denote the second-order elliptic operator B^ϵ as below

$$B^\epsilon(T) = -\frac{\partial}{\partial x_i} \left(k_{ij}^\epsilon(\mathbf{x}) \frac{\partial T}{\partial x_j} \right). \quad (4)$$

With this notation and the chain rule (2), it can be checked that

$$B^\epsilon(T) = \epsilon^{-2} B_0(T) + \epsilon^{-1} B_1(T) + \epsilon^0 B_2(T), \quad (5)$$

where

$$\begin{cases} B_0(T) = -\frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial T}{\partial y_j} \right), \\ B_1(T) = -\frac{\partial}{\partial x_i} \left(k_{ij}(\mathbf{y}) \frac{\partial T}{\partial y_j} \right) - \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial T}{\partial x_j} \right), \\ B_2(T) = -\frac{\partial}{\partial x_i} \left(k_{ij}(\mathbf{y}) \frac{\partial T}{\partial x_j} \right). \end{cases} \quad (6)$$

Then, with the help of classical multi-scale asymptotic theory [57], by substituting (3) into (1) and equating the terms with the same power of ϵ , we hence derive a series of equations as follows.

$$O(\epsilon^{-2}) : B_0(T^{(0)}) = 0, \quad (7)$$

$$O(\epsilon^{-1}) : B_0(T^{(1)}) + B_1(T^{(0)}) = 0, \quad (8)$$

$$O(\epsilon^0) : \rho(\mathbf{y})c(\mathbf{y}) \frac{\partial T^{(0)}}{\partial t} + B_0(T^{(2)}) + B_1(T^{(1)}) + B_2(T^{(0)}) = f(\mathbf{x}, t). \quad (9)$$

Owing to the periodic property of $T^{(0)}$ with variable $\mathbf{y}$, the multi-scale asymptotic theory implies that $T^{(0)}(\mathbf{x}, \mathbf{y}, t)$ is independent of variable $\mathbf{y}$, namely

$$T^{(0)}(\mathbf{x}, \mathbf{y}, t) = T^{(0)}(\mathbf{x}, t). \quad (10)$$

Furthermore, substituting (10) into (8), this leads to

$$\frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial T^{(1)}}{\partial y_j} \right) = - \frac{\partial k_{ij}(\mathbf{y})}{\partial y_i} \frac{\partial T^{(0)}}{\partial x_j}. \quad (11)$$

According to the equality (10), the detailed expression for first-order asymptotic term $T^{(1)}$ is then given by

$$T^{(1)}(\mathbf{x}, \mathbf{y}, t) = \mathcal{M}_{\alpha_1}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}}, \quad (12)$$

where $\mathcal{M}_{\alpha_1}(\mathbf{y})$ is denoted as first-order (lower-order) auxiliary UC function defined on microscopic UC Y . Now, substituting (12) into (11) and imposing the homogeneous Dirichlet boundary condition, $\mathcal{M}_{\alpha_1}(\mathbf{y})$ is defined as the solution to the following lower-order cell problem.

$$\begin{cases} \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_1}(\mathbf{y})}{\partial y_j} \right) = - \frac{\partial k_{i\alpha_1}(\mathbf{y})}{\partial y_i}, & \text{in } Y, \\ \mathcal{M}_{\alpha_1}(\mathbf{y}) = 0, & \text{on } \partial Y. \end{cases} \quad (13)$$

By combining the assumption A₁ and Lax-Milgram theorem, we can demonstrate that lower-order cell problem (13) has the unique solution $\mathcal{M}_{\alpha_1}(\mathbf{y})$.

Subsequently, taking the volume integral of both sides of (9) on UC Y and simultaneously applying Gauss theorem to (9), it follows immediately that

$$\begin{cases} S^* \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} - \frac{\partial}{\partial x_i} \left(k_{ij}^* \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_j} \right) = f(\mathbf{x}, t), & \text{in } \Omega \times (0, T^*), \\ T^{(0)}(\mathbf{x}, t) = \hat{T}(\mathbf{x}, t), & \text{on } \partial\Omega \times (0, T^*), \\ T^{(0)}(\mathbf{x}, 0) = \check{T}(\mathbf{x}), & \text{in } \Omega, \end{cases} \quad (14)$$

where macroscopic homogenized material parameters S^* and k_{ij}^* can be calculated according to the following formula.

$$\begin{aligned} S^* &= \frac{1}{|Y|} \int_Y \rho(\mathbf{y}) c(\mathbf{y}) dY, \\ k_{ij}^* &= \frac{1}{|Y|} \int_Y \left(k_{ij}(\mathbf{y}) + k_{ik}(\mathbf{y}) \frac{\partial \mathcal{M}_j(\mathbf{y})}{\partial y_k} \right) dY. \end{aligned} \quad (15)$$

In turn, the solution procedure for the second-order (higher-order) asymptotic term $T^{(2)}$ is given in detail. By subtracting equation (9) from macroscopic

homogenized equation (14) and substituting the detailed expression for $T^{(1)}$ into (9), we can derive the following equality

$$\begin{aligned} \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial T^{(2)}}{\partial y_j} \right) &= \left[k_{\alpha_1 \alpha_2}^* - k_{\alpha_1 \alpha_2}(\mathbf{y}) - k_{\alpha_1 j}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_2}(\mathbf{y})}{\partial y_j} \right. \\ &\quad \left. - \frac{\partial}{\partial y_i} (k_{i \alpha_2}(\mathbf{y}) \mathcal{M}_{\alpha_1}(\mathbf{y})) \right] \frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + \left[\rho(\mathbf{y}) c(\mathbf{y}) - S^* \right] \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t}. \end{aligned} \quad (16)$$

With the help of equality (16), the specific expression of higher-order asymptotic term $T^{(2)}$ hereafter can be defined as below.

$$T^{(2)}(\mathbf{x}, \mathbf{y}, t) = \mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + \mathcal{R}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t}, \quad (17)$$

where $\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y})$ and $\mathcal{R}(\mathbf{y})$ are denoted as second-order (higher-order) auxiliary UC functions defined on microscopic UC Y . Analogously, implementing the substitution (17) into (16) and imposing the homogeneous Dirichlet boundary condition, $\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y})$ and $\mathcal{R}(\mathbf{y})$ are denoted as the solutions to the following higher-order cell problems.

$$\begin{cases} \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y})}{\partial y_j} \right) = k_{\alpha_1 \alpha_2}^* - k_{\alpha_1 \alpha_2}(\mathbf{y}) - k_{\alpha_1 j}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_2}(\mathbf{y})}{\partial y_j} \\ \quad - \frac{\partial}{\partial y_i} (k_{i \alpha_2}(\mathbf{y}) \mathcal{M}_{\alpha_1}(\mathbf{y})), \text{ in } Y, \\ \mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}) = 0, \text{ on } \partial Y, \end{cases} \quad (18)$$

$$\begin{cases} \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \mathcal{R}(\mathbf{y})}{\partial y_j} \right) = \rho(\mathbf{y}) c(\mathbf{y}) - S^*, \text{ in } Y, \\ \mathcal{R}(\mathbf{y}) = 0, \text{ on } \partial Y. \end{cases} \quad (19)$$

Based on the assumption A₁ and Lax-Milgram theorem, we can prove that higher-order cell problems (18) and (19) have the unique solutions $\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y})$ and $\mathcal{R}(\mathbf{y})$, respectively.

Summing up, with respect to the temperature solution $T^\epsilon(\mathbf{x}, t)$, its higher-order multi-scale asymptotic solution $T^{(2\epsilon)}(\mathbf{x}, t)$ is established as below.

$$\begin{aligned} T^{(2\epsilon)}(\mathbf{x}, t) &= T^{(0)}(\mathbf{x}, t) + \epsilon \mathcal{M}_{\alpha_1}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} \\ &\quad + \epsilon^2 \left[\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + \mathcal{R}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} \right], \end{aligned} \quad (20)$$

where $T^{(0)}(\mathbf{x})$ is the macroscopic homogenized solution, which can capture the macroscopic behavior of the original multi-scale problem. $\mathcal{M}_{\alpha_1}(\mathbf{y})$ is the lower-order auxiliary UC function, which can capture the low-frequency oscillatory behavior due to spatial heterogeneity of composite materials. $\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y})$ and

$\mathcal{R}(\mathbf{y})$ are the higher-order auxiliary UC functions, which can capture the high-frequency oscillatory behavior due to spatial heterogeneity of composite materials.

To the best of our knowledge, the solution $T^\epsilon(\mathbf{x}, t; \boldsymbol{\theta}^*)$ of neural network-based approach is determined by minimizing the loss function (21), over the admissible set of tuning parameters $\boldsymbol{\theta}^* \in \mathbb{R}^{\mathcal{D}}$

$$\left\{ \begin{array}{l} \text{Find: } T^\epsilon(\mathbf{x}, t; \boldsymbol{\theta}^*) \text{ with } \boldsymbol{\theta}^* \in \mathbb{R}^{\mathcal{D}}, \\ \text{s.t. } \boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^{\mathcal{D}}} \left\{ \lambda_G \left\| \rho^\epsilon(\mathbf{x}) c^\epsilon(\mathbf{x}) \frac{\partial T^\epsilon(\mathbf{x}, t; \boldsymbol{\theta})}{\partial t} - \frac{\partial}{\partial x_i} (k_{ij}^\epsilon(\mathbf{x}) \frac{\partial T^\epsilon(\mathbf{x}, t; \boldsymbol{\theta})}{\partial x_j}) \right. \right. \\ \left. \left. - f(\mathbf{x}, t) \right\|_{\Omega \times (0, T^*)} + \lambda_B \left\| T^\epsilon(\mathbf{x}, t; \boldsymbol{\theta}) - \hat{T}(\mathbf{x}, t) \right\|_{\partial \Omega \times (0, T^*)} + \lambda_I \left\| T^\epsilon(\mathbf{x}, 0; \boldsymbol{\theta}) - \check{T}(\mathbf{x}) \right\|_{\Omega} \right\}, \end{array} \right. \quad (21)$$

where $\|\bullet\|_{\Omega \times (0, T^*)}$, $\|\bullet\|_{\partial \Omega \times (0, T^*)}$ and $\|\bullet\|_{\Omega}$ represent different norms in Sobolev space, which will be used in the sequel. Moreover, λ_G denotes the penalty or weight coefficients for the governing equations; λ_B denotes the penalty or weight coefficients for the boundary condition of PDEs; λ_I denotes the penalty or weight coefficient for the initial condition of PDEs.

On the basis of the above-mentioned higher-order multi-scale model, original multi-scale solution $T^\epsilon(\mathbf{x})$ with highly oscillatory characteristic of multi-scale PDE (1) is accurately approximated by higher-order multi-scale solution $T^{(2\epsilon)}(\mathbf{x}, t)$. Therefore, new physical constrains are designed and established for solving multi-scale parabolic PDE (1) by neural network-based approaches. After that, we can define the novel higher-order multi-scale solution $T_{\text{NN}}^{(2\epsilon)}(\mathbf{x}, t)$ based on neural network-based method for multi-scale heat conduction problem (1) of composite materials as below.

$$\begin{aligned} T_{\text{NN}}^{(2\epsilon)}(\mathbf{x}, t) = & T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta}_0^*) + \epsilon \mathcal{M}_{\alpha_1}(\mathbf{y}; \boldsymbol{\theta}_1^*) \frac{\partial T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta}_0^*)}{\partial x_{\alpha_1}} \\ & + \epsilon^2 [\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}; \boldsymbol{\theta}_2^*) \frac{\partial^2 T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta}_0^*)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + \mathcal{R}(\mathbf{y}; \boldsymbol{\theta}_3^*) \frac{\partial T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta}_0^*)}{\partial t}], \end{aligned} \quad (22)$$

where $T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta}_0^*)$, $\mathcal{M}_{\alpha_1}(\mathbf{y}; \boldsymbol{\theta}_1^*)$, $\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}; \boldsymbol{\theta}_2^*)$ and $\mathcal{R}(\mathbf{y}; \boldsymbol{\theta}_3^*)$ are determined by solving the following optimization problems respectively.

$$\left\{ \begin{array}{l} \text{Find: } T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta}_0^*) \text{ with } \boldsymbol{\theta}_0^* \in \mathbb{R}^{\mathcal{D}}, \\ \text{s.t. } \boldsymbol{\theta}_0^* = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^{\mathcal{D}}} \left\{ \lambda_{G_0} \left\| S^* \frac{\partial T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta})}{\partial t} - \frac{\partial}{\partial x_i} (k_{ij}^*(\mathbf{x}) \frac{\partial T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta})}{\partial x_j}) \right. \right. \\ \left. \left. - f(\mathbf{x}, t) \right\|_{\Omega \times (0, T^*)} + \lambda_{B_0} \left\| T^{(0)}(\mathbf{x}, t; \boldsymbol{\theta}) - \hat{T}(\mathbf{x}, t) \right\|_{\partial \Omega \times (0, T^*)} + \lambda_{I_0} \left\| T^{(0)}(\mathbf{x}, 0; \boldsymbol{\theta}) - \check{T}(\mathbf{x}) \right\|_{\Omega} \right\}, \end{array} \right. \quad (23)$$

$$\begin{cases} \text{Find: } \mathcal{M}_{\alpha_1}(\mathbf{y}; \boldsymbol{\theta}_1^*) \text{ with } \boldsymbol{\theta}_1^* \in \mathbb{R}^{\mathcal{D}}, \\ \text{s.t. } \boldsymbol{\theta}_1^* = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^{\mathcal{D}}} \left\{ \lambda_{G_1} \left\| \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_1}(\mathbf{y}; \boldsymbol{\theta})}{\partial y_j} \right) + \frac{\partial k_{i\alpha_1}(\mathbf{y})}{\partial y_i} \right\|_Y \right. \\ \left. + \lambda_{B_1} \left\| \mathcal{M}_{\alpha_1}(\mathbf{y}; \boldsymbol{\theta}) \right\|_{\partial Y} \right\}, \end{cases} \quad (24)$$

$$\begin{cases} \text{Find: } \mathcal{M}_{\alpha_1\alpha_2}(\mathbf{y}; \boldsymbol{\theta}_2^*) \text{ with } \boldsymbol{\theta}_2^* \in \mathbb{R}^{\mathcal{D}}, \\ \text{s.t. } \boldsymbol{\theta}_2^* = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^{\mathcal{D}}} \left\{ \lambda_{G_2} \left\| \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_1\alpha_2}(\mathbf{y}; \boldsymbol{\theta})}{\partial y_j} \right) - k_{\alpha_1\alpha_2}^* + k_{\alpha_1\alpha_2}(\mathbf{y}) \right. \right. \\ \left. \left. + k_{\alpha_1j}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_2}(\mathbf{y}; \boldsymbol{\theta}_1^*)}{\partial y_j} + \frac{\partial}{\partial y_i} (k_{i\alpha_2}(\mathbf{y}) \mathcal{M}_{\alpha_1}(\mathbf{y}; \boldsymbol{\theta}_1^*)) \right\|_Y + \lambda_{B_2} \left\| \mathcal{M}_{\alpha_1\alpha_2}(\mathbf{y}; \boldsymbol{\theta}) \right\|_{\partial Y} \right\}, \end{cases} \quad (25)$$

$$\begin{cases} \text{Find: } \mathcal{R}(\mathbf{y}; \boldsymbol{\theta}_3^*) \text{ with } \boldsymbol{\theta}_3^* \in \mathbb{R}^{\mathcal{D}}, \\ \text{s.t. } \boldsymbol{\theta}_3^* = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^{\mathcal{D}}} \left\{ \lambda_{G_3} \left\| \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \mathcal{R}(\mathbf{y}; \boldsymbol{\theta})}{\partial y_j} \right) - \rho(\mathbf{y})c(\mathbf{y}) + S^* \right\|_Y \right. \\ \left. + \lambda_{B_3} \left\| \mathcal{R}(\mathbf{y}; \boldsymbol{\theta}) \right\|_{\partial Y} \right\}. \end{cases} \quad (26)$$

Among the above neural network-based models (23)-(26), λ_{G_0} , λ_{G_1} , λ_{G_2} and λ_{G_3} denote the penalty or weight coefficients for the governing equations; λ_{B_0} , λ_{B_1} , λ_{B_2} and λ_{B_3} denote the penalty or weight coefficients for the boundary condition of PDEs; λ_{I_0} denotes the penalty or weight coefficient for the initial condition of PDEs.

3. Higher-order multi-scale randomized neural network method

Through the higher-order multi-scale model of heterogeneous materials in Section 2, we decouple original physical constraints (multi-scale PDEs) of neural networks as new physical constraints (macroscopic homogenized PDEs and microscopic cell PDEs) of neural networks, which resolve the prohibitive computation and Frequency Principle, and possess the high-accuracy computational performance for multi-scale thermal simulation. In this section, we firstly briefly introduce randomized neural network, and then design two kinds of randomized neural networks to solve microscopic auxiliary UC problems and macroscopic homogenized problem. Moreover, by introducing hard-constraint boundary condition to assist RNN method, the computational accuracy of RNN method substantially improve for solving elliptic type microscopic cell problems of high-contrast inhomogeneous materials. After that, the algorithmic procedures of the proposed HOMS-RNN framework are described in detail.

3.1. Randomized neural networks (RNNs)

In this subsection, we briefly introduce the principal idea and some concepts of RNN. To the best of our knowledge, a fully connected neural network $\mathcal{N}(\mathbf{x}; \boldsymbol{\theta}) : \mathbb{R}^{n_0} \rightarrow \mathbb{R}^{n_D}$ with depth D can be defined as follows.

$$\begin{aligned}\Phi_0(\mathbf{x}) &= \mathbf{x}, \\ \Phi_l(\mathbf{x}) &= \sigma(\mathbf{W}_l \Phi_{l-1} + \mathbf{b}_l), \quad l = 1, \dots, D-1, \\ \mathcal{N}(\mathbf{x}; \boldsymbol{\theta}) &= \Phi_D(\mathbf{x}) = \mathbf{W}_D \Phi_{D-1} + \mathbf{b}_D,\end{aligned}\tag{27}$$

where $\sigma(\bullet)$ denotes the activation function, $\mathbf{W}_l \in \mathbb{R}^{n_l \times \mathbb{R}^{n_{l-1}}}$ and $\mathbf{b}_l \in \mathbb{R}^{n_l}$ denote the weights and the bias in the l -th layer, $\boldsymbol{\theta} = [(\mathbf{W}_1, \mathbf{b}_1); \dots; (\mathbf{W}_D, \mathbf{b}_D)]$.

Randomized neural network (RNN) or extreme learning machine (ELM) is a special type of deep neural networks, whose prominent feature is that randomly selecting and fixing all biases and weights $[(\mathbf{W}_1, \mathbf{b}_1); \dots; (\mathbf{W}_{D-1}, \mathbf{b}_{D-1})]$ (non-trainable) except for those in the last layer, other weights $\mathbf{W}_D$ of the last layer are then computed by solving a least-square problem without gradient descent-type optimization solvers. In the current work, it should mention that the D -th layer has zero bias, i.e. $\mathbf{b}_D = \mathbf{0}$ for randomized neural networks. Besides, the coefficients $[(\mathbf{W}_1, \mathbf{b}_1); \dots; (\mathbf{W}_{D-1}, \mathbf{b}_{D-1})]$ will be assigned to uniform random values from $[-R_m, R_m]$ and fixed.

Denote that $\mathcal{N}_\sigma^D$ is the set of functions that can be represented by RNNs, and $\{\phi_1, \dots, \phi_{n_{D-1}}\}$ is a basis for this space, where ϕ_i is the i -th component of the vector Φ_{D-1} and is fixed once the hidden-layer coefficients are randomly assigned. Therefore, any function $u_\sigma \in \mathcal{N}_\sigma^D$ (one neuron in the output layer) can be expressed as the following linear combination of the basis functions, where $\mathbf{W}_D = (\alpha_1, \dots, \alpha_{D-1})^T$ denote the output-layer coefficients (trainable parameters) of RNNs.

$$u_\sigma(\mathbf{x}) = \sum_{i=1}^{n_{D-1}} \alpha_i \phi_i(\mathbf{x}).\tag{28}$$

Moreover, the network structure of a multi-hidden layers RNN is exhibited in Fig. 2.

Inspired by the universal approximation theorems in [32, 58], it is basically known that any given continuous function can be approximated by a randomized neural network possessing a single hidden layer, in which the hidden-layer coefficients are randomly assigned and the output-layer coefficients can be adjusted/trained to any desired degree of accuracy, as long as the number of hidden neurons is adequately large. Hence, RNNs can be effectively leveraged to assist scientific computing for PDEs.

3.2. RNNs for solving microscopic and macroscopic physical constrains

In subsection 2.2, neural network-based models for microscopic and macroscopic physical constrains of multi-scale PDE (1) of composite materials are established as (23)-(26). In virtue of excellent approximation performance and efficient computation capability, RNNs are employed to solve microscopic and

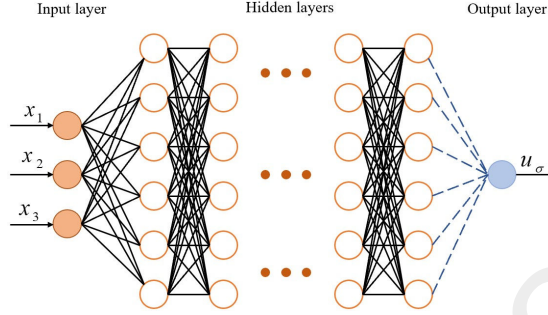


Figure 2: Network structure of $u_\sigma : \mathbb{R}^3 \rightarrow \mathbb{R}$, the solid black lines refer to the parameters of u_σ that are randomly initialized and fixed thereafter, while the dotted blue lines represent to the parameters that require to be solved.

macroscopic physical constrains (23)-(26) of multi-scale PDE (1) of composite materials.

Firstly, we denote two RNNs $u_\sigma^1(\mathbf{y})$ and $\tilde{u}_\sigma^1(\mathbf{y})$ for approximating microscopic cell function $\mathcal{M}_{\alpha_1}(\mathbf{y})$ in matrix material Y_1 and inclusion material Y_2 as below.

$$u_\sigma^1(\mathbf{y}) = \sum_{m=1}^{M_1} \alpha_m^1 \phi_m^1(\mathbf{y}), \quad \tilde{u}_\sigma^1(\mathbf{y}) = \sum_{m=1}^{M_2} \tilde{\alpha}_m^1 \tilde{\phi}_m^1(\mathbf{y}). \quad (29)$$

Then, we substitute (29) into the microscopic cell problem (13) of microscopic cell function $\mathcal{M}_{\alpha_1}(\mathbf{y})$, and discretize the problem at randomly sampled N_1 , N_2 and N_3 collocation points on Y , ∂Y and $\Gamma = Y_1 \cap Y_2$ respectively, then we can obtain a linear system as below.

$$\begin{cases} \mathbf{A}\mathbf{X} = \mathbf{F}, & \text{in } Y, \\ \lambda_1 \mathbf{B}\mathbf{X} = \lambda_1 \cdot \mathbf{0}, & \text{on } \partial Y, \\ \lambda_2 \mathbf{C}\mathbf{X} = \lambda_2 \cdot \mathbf{0}, & \text{on } \Gamma = Y_1 \cap Y_2, \\ \lambda_3 \mathbf{D}\mathbf{X} = \lambda_3 \cdot \mathbf{0}, & \text{on } \Gamma = Y_1 \cap Y_2, \end{cases} \quad (30)$$

where $\mathbf{A}$ and $\mathbf{B}$ are matrices of order $N_1 \times (M_1 + M_2)$ and $N_2 \times (M_1 + M_2)$, respectively. $\mathbf{C}$ and $\mathbf{D}$ are matrices of order $N_3 \times (M_1 + M_2)$ and $N_3 \times (M_1 + M_2)$, respectively. $\mathbf{F}$ is a vector of order N_1 . $\mathbf{X}$ is the unsolved vector of order $M_1 + M_2$. Moreover, λ_1 , λ_2 and λ_3 represent the adjustable weights in the linear system to measure the importance of different conditions and control the

attention of RNNs. Specifically, they have the detailed expressions as follows.

$$\left\{ \begin{array}{l} A_{n,m} = \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \phi_m^1(\mathbf{y}_n)}{\partial y_j} \right), \quad \mathbf{y}_n \in Y_1, 1 \leq m \leq M_1, \\ A_{n,m} = \frac{\partial}{\partial y_i} \left(k_{ij}(\mathbf{y}) \frac{\partial \tilde{\phi}_{m-M_1}^1(\mathbf{y}_n)}{\partial y_j} \right), \quad \mathbf{y}_n \in Y_2, M_1 + 1 \leq m \leq M_1 + M_2, \\ F_n = -\frac{\partial k_{i\alpha_1}(\mathbf{y}_n)}{\partial y_i}, \quad \mathbf{y}_n \in Y, \\ B_{n,m} = \phi_m^1(\mathbf{y}_n), \quad \mathbf{y}_n \in \partial Y, 1 \leq m \leq M_1, \\ B_{n,m} = 0, \quad \mathbf{y}_n \in \partial Y, M_1 + 1 \leq m \leq M_1 + M_2, \\ C_{n,m} = \phi_m^1(\mathbf{y}_n), \quad \mathbf{y}_n \in \Gamma, 1 \leq m \leq M_1, \\ C_{n,m} = -\tilde{\phi}_{m-M_1}^1(\mathbf{y}_n), \quad \mathbf{y}_n \in \Gamma, M_1 + 1 \leq m \leq M_1 + M_2, \\ D_{n,m} = k_{ij}(\mathbf{y}) \frac{\partial \phi_m^1(\mathbf{y}_n)}{\partial y_j} n_i(\mathbf{y}_n), \quad \mathbf{y}_n \in \Gamma, 1 \leq m \leq M_1, \\ D_{n,m} = -k_{ij}(\mathbf{y}) \frac{\partial \tilde{\phi}_{m-M_1}^1(\mathbf{y}_n)}{\partial y_j} n_i(\mathbf{y}_n), \quad \mathbf{y}_n \in \Gamma, M_1 + 1 \leq m \leq M_1 + M_2, \\ \mathbf{X} = (\alpha_1^1, \dots, \alpha_{M_1}^1, \tilde{\alpha}_1^1, \dots, \tilde{\alpha}_{M_2}^1)^T. \end{array} \right. \quad (31)$$

Based on the foregoing analysis and computation, microscopic cell function $\mathcal{M}_{\alpha_1}(\mathbf{y})$ can be computed by solving linear system (30) of RNNs, rather than the optimization problem (24) through gradient descent-type algorithms. In order to compute the matrices $\mathbf{A}$ and $\mathbf{D}$, and vector $\mathbf{F}$, some involving derivatives should be accurately and efficiently evaluated. In this study, finite difference schemes [34, 35] are employed to replace the automatic differentiation approach of neural networks for computing involving derivatives since it is inefficient. Besides, smooth approximation technique is employed to compute the vector $\mathbf{F}$, which is formed only by the material constants. As for authentic composite materials, their material parameters are highly discontinuous, which means that F_n is zero value at matrix and inclusion, and non-zero concentrated load distributed along interface lines of matrix and inclusion. Hence, the continuous functions is established to smoothly approximate material parameters and tackle the non-differentiable issue at interface. More specifically, for 2D composite material with circular inclusions as shown in Fig. 1, its smooth approximation function is presented as follows.

$$f_k = (c_1 - c_2) f_s \left(k \left((y_1 - o_1)^2 + (y_2 - o_2)^2 - r^2 \right) \right) + c_2, \quad (32)$$

where c_1 and c_2 represents the material parameters of matrix and inclusion, f_s represents sigmoid function, k indicates the smoothing parameter of material transition, (o_1, o_2) denotes the central coordinates of circular inclusion, and r denotes the radius of circular inclusion, as illustrated in Fig. 3. Finally, by solving the least-square problem with the linear system (30), the weight vector $\mathbf{X}$ of the output layer for the RNNs $u_\sigma^1(\mathbf{y})$ and $\tilde{u}_\sigma^1(\mathbf{y})$ is obtained to numerically

approximate $\mathcal{M}_{\alpha_1}(\mathbf{y})$.

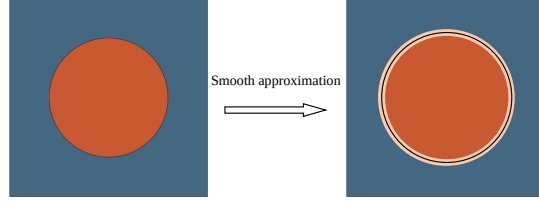


Figure 3: The smooth approximation of material parameters.

Additionally, to further improve the numerical accuracy of RNN method, hard-constrained boundary conditions [59, 60] are employed to modify the original RNN and resolve the loss imbalance problem for microscopic cell function $\mathcal{M}_{\alpha_1}(\mathbf{y})$. The surrogate model of RNN $u_\sigma^1(\mathbf{y})$ for solving $\mathcal{M}_{\alpha_1}(\mathbf{y})$ can be given as below.

$$\hat{u}_\sigma^1(\mathbf{y}) = h(\mathbf{y}) + \ell(\mathbf{y})u_\sigma^1(\mathbf{y}), \quad (33)$$

where $h(\mathbf{y}) = 0$ for homogeneous Dirichlet boundary condition of $\mathcal{M}_{\alpha_1}(\mathbf{y})$ and $\ell(\mathbf{y})$ is a function that satisfies the following constrained condition.

$$\begin{cases} \ell(\mathbf{y}) > 0, & \text{in } Y, \\ \ell(\mathbf{y}) = 0, & \text{on } \partial Y. \end{cases} \quad (34)$$

For example, we can define $\ell(\mathbf{y})$ as function $y_1 y_2 (1 - y_1) (1 - y_2)$ for 2D microscopic unit cell $[0, 1]^2$ in 2D case.

Furthermore, we define one space-time RNN $u_\sigma^0(\mathbf{x}, t)$ for approximating macroscopic homogenized solution $T^{(0)}(\mathbf{x}, t)$ as below.

$$u_\sigma^0(\mathbf{x}, t) = \sum_{m=1}^M \alpha_m^0 \phi_m^0(\mathbf{x}, t). \quad (35)$$

Analogously, substituting (35) into the macroscopic homogenized problem (14), and discretizing spatial-domain Ω , boundary-domain $\partial\Omega$ and temporal-domain $(0, T^*)$ with randomly sampled $\hat{N}_1$, $\hat{N}_2$ and $\hat{N}_3$ collocation points respectively, then we can obtain the following linear system for macroscopic homogenized solution $T^{(0)}(\mathbf{x}, t)$.

$$\begin{cases} \hat{\mathbf{A}}\hat{\mathbf{X}} = \hat{\mathbf{F}}, & \text{in } \Omega \times (0, T^*), \\ \gamma_1 \hat{\mathbf{B}}\hat{\mathbf{X}} = \gamma_1 \hat{\mathbf{G}}, & \text{on } \partial\Omega \times (0, T^*), \\ \gamma_2 \hat{\mathbf{C}}\hat{\mathbf{X}} = \gamma_2 \hat{\mathbf{H}}, & \text{on } \Omega, \end{cases} \quad (36)$$

where $\hat{\mathbf{A}}$, $\hat{\mathbf{B}}$ and $\hat{\mathbf{C}}$ are matrices of order $(\hat{N}_1 \times \hat{N}_3) \times M$, $(\hat{N}_2 \times \hat{N}_3) \times M$ and $\hat{N}_1 \times M$, respectively. $\hat{\mathbf{F}}$, $\hat{\mathbf{G}}$ and $\hat{\mathbf{H}}$ are vectors of order $\hat{N}_1 \times \hat{N}_3$, $\hat{N}_2 \times \hat{N}_3$ and

$\hat{N}_1$, respectively. $\hat{\mathbf{X}}$ is the unsolved vector of order M . Moreover, γ_1 and γ_2 represent the adjustable weights in the linear system of RNN. Specifically,

$$\begin{cases} \hat{A}_{n_1 \times n_3, m} = S^* \frac{\partial \phi_m^0(\mathbf{x}_{n_1}, t_{n_3})}{\partial t} - \frac{\partial}{\partial x_i} (k_{ij}^* \frac{\partial \phi_m^0(\mathbf{x}_{n_1}, t_{n_3})}{\partial x_j}), & \mathbf{x}_{n_1} \in \Omega, t_{n_3} \in (0, T^*), \\ \hat{F}_{n_1 \times n_3} = f(\mathbf{x}_{n_1}, t_{n_3}), & \mathbf{x}_{n_1} \in \Omega, t_{n_3} \in (0, T^*), \\ \hat{B}_{n_2 \times n_3, m} = \phi_m^0(\mathbf{x}_{n_2}, t_{n_3}), & \mathbf{x}_{n_2} \in \partial\Omega, t_{n_3} \in (0, T^*), \\ \hat{G}_{n_2 \times n_3} = \hat{T}(\mathbf{x}_{n_2}, t_{n_3}), & \mathbf{x}_{n_2} \in \partial\Omega, t_{n_3} \in (0, T^*), \\ \hat{C}_{n_1, m} = \phi_m^0(\mathbf{x}_{n_1}, 0), & \mathbf{x}_{n_1} \in \Omega, \\ \hat{H}_{n_1} = \check{T}(\mathbf{x}_{n_1}), & \mathbf{x}_{n_1} \in \Omega, \\ \hat{\mathbf{X}} = (\alpha_1^0, \dots, \alpha_M^0)^T. \end{cases} \quad (37)$$

This linear system can be space-time solved via the least-square method.

Furthermore, the surrogate model of RNN $u_\sigma^0(\mathbf{x}, t)$ for solving $T^{(0)}(\mathbf{x}, t)$ can be presented as below.

$$\hat{u}_\sigma^0(\mathbf{x}, t) = h(\mathbf{x}) + \ell(\mathbf{x})u_\sigma^0(\mathbf{x}, t), \quad (38)$$

where $h(\mathbf{x}) = 0$ for homogeneous Dirichlet boundary condition of $T^{(0)}(\mathbf{x}, t)$ and $\ell(\mathbf{x})$ is a function that satisfies the following constrained condition.

$$\begin{cases} \ell(\mathbf{x}) > 0, & \text{in } \Omega, \\ \ell(\mathbf{x}) = 0, & \text{on } \partial\Omega. \end{cases} \quad (39)$$

For example, we can define $\ell(\mathbf{x})$ as function $x_1 x_2 (1 - x_1) (1 - x_2)$ for 2D macroscopic homogenized material $[0, 1]^2$ in 2D case.

After that, we denote two RNNs $u_\sigma^2(\mathbf{y})$ and $\tilde{u}_\sigma^2(\mathbf{y})$ for approximating microscopic cell function $\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y})$ in matrix material Y_1 and inclusion material Y_2 as below.

$$u_\sigma^2(\mathbf{y}) = \sum_{m=1}^{M_1} \alpha_m^2 \phi_m^2(\mathbf{y}), \quad \tilde{u}_\sigma^2(\mathbf{y}) = \sum_{m=1}^{M_2} \tilde{\alpha}_m^2 \tilde{\phi}_m^2(\mathbf{y}). \quad (40)$$

Then, we substitute (40) into the microscopic cell problem (18) of microscopic cell function $\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y})$, and discretize the problem as a linear system, which can be solved by the similar method to linear system (30).

Furthermore, we denote two RNNs $u_\sigma^3(\mathbf{y})$ and $\tilde{u}_\sigma^3(\mathbf{y})$ for approximating microscopic cell function $\mathcal{R}(\mathbf{y})$ in matrix material Y_1 and inclusion material Y_2 as below.

$$u_\sigma^3(\mathbf{y}) = \sum_{m=1}^{M_1} \alpha_m^3 \phi_m^3(\mathbf{y}), \quad \tilde{u}_\sigma^3(\mathbf{y}) = \sum_{m=1}^{M_2} \tilde{\alpha}_m^3 \tilde{\phi}_m^3(\mathbf{y}). \quad (41)$$

And also, substituting (41) into the microscopic cell problem (19) of microscopic cell function $\mathcal{R}(\mathbf{y})$, and discretizing this problem as a linear system, then it can be solved by the similar method to linear system (30).

To summarize, on the basis of the above-mentioned RNN computation, the high-accuracy numerical solutions of $\mathcal{M}_{\alpha_1}(\mathbf{y})$, $T^{(0)}(\mathbf{x}, t)$, $\mathcal{M}_{\alpha_1\alpha_2}(\mathbf{y})$ and $\mathcal{R}(\mathbf{y})$ are obtained. Finally, according to higher-order multi-scale asymptotic formulas (20) and (22), and using automatic differentiation approach, the high-accuracy numerical solution (42) for multi-scale heat conduction problem (1) is accurately and efficiently solved by RNN approach for composite materials.

$$\begin{aligned} \bar{T}^{(2\epsilon)}(\mathbf{x}, t) = & \bar{T}^{(0)}(\mathbf{x}, t) + \epsilon \bar{\mathcal{M}}_{\alpha_1}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} \\ & + \epsilon^2 \left[\bar{\mathcal{M}}_{\alpha_1\alpha_2}(\mathbf{y}) \frac{\partial^2 \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + \bar{\mathcal{R}}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial t} \right], \end{aligned} \quad (42)$$

where $\bar{T}^{(0)}(\mathbf{x}, t)$, $\bar{\mathcal{M}}_{\alpha_1}(\mathbf{y})$, $\bar{\mathcal{M}}_{\alpha_1\alpha_2}(\mathbf{y})$ and $\bar{\mathcal{R}}(\mathbf{y})$ are determined by RNN approach respectively.

3.3. Algorithmic procedures of HOMS-RNN method

In this subsection, we firstly present the detailed theoretical idea of the proposed HOMS-RNN approach. The proposed HOMS-RNN approach inherits the pivotal strengths of the higher-order multi-scale method and RNN approach. In the HOMS-RNN framework, higher-order multi-scale approach is employed to address the large complexity and Frequency Principle, while RNN approach is utilized to efficiently and mesh-free solve higher-order multi-scale computational model, including lower-order and higher-order microscopic cell functions, and macroscopic homogenized equation without multi-scale feature. From the Frequency Principle perspective, the presented HOMS-RNN approach decomposes the highly oscillatory solution of multi-scale PDE into the combination of non-oscillatory information in macroscopic homogenized solution, low-frequency oscillatory information in lower-order microscopic cell functions and high-frequency oscillatory information in higher-order microscopic cell functions, as illustrated in Fig. 4(a). From the numerical computation perspective, direct simulation of neural network-based approach requires a tremendous amount of sampling residual points to capture the locally oscillatory response of inhomogeneous materials due to highly spatial heterogeneities, which leads to prohibitive expend of computational resource and computational challenge of neural network convergence. The proposed HOMS-RNN approach can mesh-free solve the macroscopic homogenized problem and microscopic cell problems with a small quantity of residual points respectively, that can effectively save computational resource and dramatically enhance the trainable efficiency of RNNs. Moreover, automatic differentiation technique of neural networks is applied to assemble higher-order multi-scale solution of heat conduction problem of inhomogeneous materials without the accuracy loss.

In a summary, the presented HOMS-RNN approach can efficiently simulate the heat conduction problems of inhomogeneous materials and accurately catch the microscopic oscillatory behavior of composite materials. On the basis of the

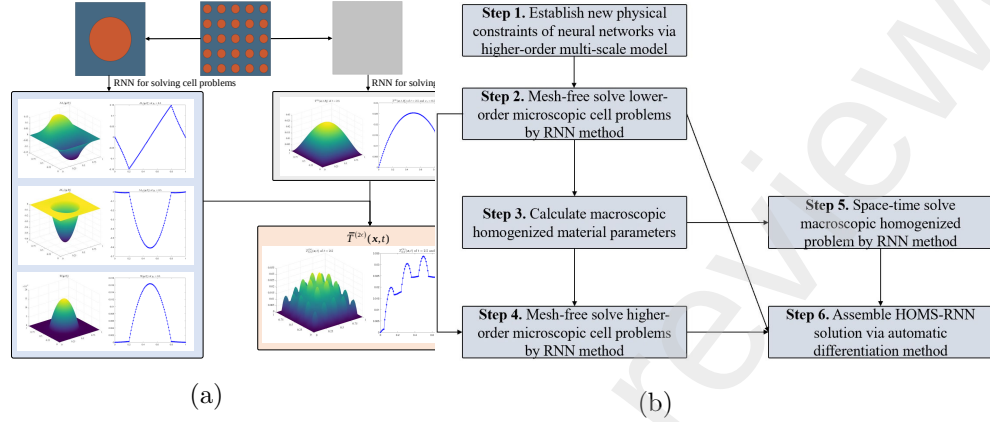


Figure 4: The illustration of HOMS-RNN approach. (a) Theoretical idea. (b) Algorithmic procedures.

pivotal mechanism of HOMS-RNN methodology, the detailed algorithm procedures, as exhibited Fig. 4(b), are presented as follows.

-
- Step 1: Decouple original physical constraints (multi-scale PDEs) of neural networks as new physical constraints (macroscopic homogenized PDEs and microscopic cell PDEs) of neural networks via the higher-order multi-scale model.
- Step 2: Discretize the lower-order microscopic cell problem (13) as linear system (30) by two RNNs $u_\sigma^1(\mathbf{y})$ and $\tilde{u}_\sigma^1(\mathbf{y})$ in (29), and then mesh-free solve the lower-order microscopic cell functions $\mathbf{M}_{\alpha_1}(\mathbf{y})$ with least-square method.
- Step 3: Employ the obtained $\mathbf{M}_{\alpha_1}(\mathbf{y})$ in Step 2, and evaluate the macroscopic homogenized material parameters S^* and k_{ij}^* by making integral of formula (15).
- Step 4: Discretize the macroscopic homogenized problem (14) as linear system (36) by space-time RNN $u_\sigma^0(\mathbf{x}, t)$ in (35), and then space-time solve the macroscopic homogenized solution $T^{(0)}(\mathbf{x}, t)$.
- Step 5: Discretize the higher-order microscopic cell problems (18) and (19) into corresponding linear systems by two RNNs $u_\sigma^2(\mathbf{y})$ and $\tilde{u}_\sigma^2(\mathbf{y})$ in (40), and two RNNs $u_\sigma^3(\mathbf{y})$ and $\tilde{u}_\sigma^3(\mathbf{y})$ in (41) respectively. Then mesh-free solve the higher-order microscopic cell functions $\mathbf{M}_{\alpha_1\alpha_2}(\mathbf{y})$ and $\mathbf{R}(\mathbf{y})$ with least-square method.
- Step 6: Utilize automatic differentiation method of neural networks to assemble higher-order multi-scale asymptotic solution $\bar{T}^{(2\epsilon)}(\mathbf{x}, t)$ according to formula (42).
-

4. Convergence analysis of HOMS-RNN method

This section now turns to the error analysis of HOMS-RNN approach. The computing errors of the proposed HOMS-RNN approach in this study chiefly arise from the modeling error of higher-order multi-scale model, and the approximate error of RNNs for computing microscopic cell functions and macroscopic homogenized equation. In reference [61], authors prove that traditional neural networks can approximate the solution well with appropriate depth and width. In references [32, 35, 38, 58], authors basically state RNNs are universal function approximators and have the universal approximation capability, namely a randomized neural network with one hidden layer can approximate any continuous function on a compact domain, as long as the hidden layer size is large enough. In references [62, 63, 64], the error analysis of higher-order multi-scale method is obtained for parabolic equations of composite materials. On the basis of these existing findings, we can establish the convergence analysis of HOMS-RNN method.

Lemma 1. *Based on universal approximation theorem in references [32, 35, 38, 58, 61, 65], for any $u \in C(I^d)$ with $I^d = [a, b]^d$ ($d = 1, 2, 3, \dots$) and $\forall \varepsilon > 0$, there exists a function $u_\sigma \in \mathcal{N}_\sigma^D$, as long as the hidden layer size is large enough, which is defined on any compact K of $C(I^d)$. Then it follows that*

$$\sup_{\mathbf{x} \in K} |u_\sigma(\mathbf{x}) - u(\mathbf{x})| < \varepsilon. \quad (43)$$

Assumption 1. *For any $u \in C^3(I^d)$ and $\forall \varepsilon > 0$, there exists a function $u_\sigma \in \mathcal{N}_\sigma^D$, as long as the hidden layer size is large enough, which is defined on any compact K of $C^3(I^d)$. Then we have*

$$\max_{a \leq 3} \sup_{\mathbf{x} \in K} |\partial_x^a u_\sigma(\mathbf{x}) - \partial_x^a u(\mathbf{x})| < \varepsilon. \quad (44)$$

Lemma 2. *According to the results in references [62, 63, 64], the error estimation for higher-order multi-scale solution $T^{(2\epsilon)}(\mathbf{x}, t)$ and exact solution $T^\epsilon(\mathbf{x}, t)$ of multi-scale parabolic problem (1) is obtained as follows.*

$$\begin{aligned} & \sup_{0 \leq t \leq T^*} \int_{\Omega} (T^\epsilon(\mathbf{x}, t) - T^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega \\ & + \int_0^{T^*} \|T^\epsilon(\mathbf{x}, t) - T^{(2\epsilon)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt \leq C(T^*)\epsilon^2, \end{aligned} \quad (45)$$

where $C(T^*)$ is a positive constant irrespective of ξ , but dependent of total time T^* and Ω .

Theorem 1. *Based on the above lemmas 1-2 and assumption 1, the following global error estimation is derived for HOMS-RNN method*

$$\begin{aligned}
& \sup_{0 \leq t \leq T^*} \int_{\Omega} (T^\epsilon(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega \\
& + \int_0^{T^*} \|T^\epsilon(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt \leq C(T^*)\epsilon^2 + C(T^*)\varepsilon^2,
\end{aligned} \tag{46}$$

where C is a positive constant irrespective of ϵ and ε , but dependent of total time T^* and Ω .

Proof : Firstly, by virtue of the triangle inequality, mean value inequality and lemma 2, it follows that

$$\begin{aligned}
& \sup_{0 \leq t \leq T^*} \int_{\Omega} (T^\epsilon(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega + \int_0^{T^*} \|T^\epsilon(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt \\
& \leq \sup_{0 \leq t \leq T^*} \int_{\Omega} 2(T^\epsilon(\mathbf{x}, t) - T^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega + \sup_{0 \leq t \leq T^*} \int_{\Omega} 2(T^{(2\epsilon)}(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega \\
& + \int_0^{T^*} 2\|T^\epsilon(\mathbf{x}, t) - T^{(2\epsilon)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt + \int_0^{T^*} 2\|T^{(2\epsilon)}(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt \\
& \leq C(T^*)\epsilon^2 + \sup_{0 \leq t \leq T^*} \int_{\Omega} 2(T^{(2\epsilon)}(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega + \int_0^{T^*} 2\|T^{(2\epsilon)}(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt.
\end{aligned} \tag{47}$$

Next, on the basis of mean value inequality, the following inequality holds

$$\begin{aligned}
& \sup_{0 \leq t \leq T^*} \int_{\Omega} (T^{(2\epsilon)}(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega \\
& \leq \sup_{0 \leq t \leq T^*} \int_{\Omega} 4(T^{(0)}(\mathbf{x}, t) - \bar{T}^{(0)}(\mathbf{x}, t))^2 d\Omega \\
& + \sup_{0 \leq t \leq T^*} \int_{\Omega} 4(\epsilon \mathcal{M}_{\alpha_1}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} - \epsilon \bar{\mathcal{M}}_{\alpha_1}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}})^2 d\Omega \\
& + \sup_{0 \leq t \leq T^*} \int_{\Omega} 4(\epsilon^2 \mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} - \epsilon^2 \bar{\mathcal{M}}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}})^2 d\Omega \\
& + \sup_{0 \leq t \leq T^*} \int_{\Omega} 4(\epsilon^2 \mathcal{R}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} - \epsilon^2 \bar{\mathcal{R}}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial t})^2 d\Omega.
\end{aligned} \tag{48}$$

Moreover, by means of the triangle inequality, lemma 1 and assumption 1, we can obtain the error estimations for right side of the inequality (48) as below.

$$\sup_{0 \leq t \leq T^*} \int_{\Omega} 4(T^{(0)}(\mathbf{x}, t) - \bar{T}^{(0)}(\mathbf{x}, t))^2 d\Omega \leq C\varepsilon^2, \tag{49}$$

$$\begin{aligned}
& \sup_{0 \leq t \leq T^*} \int_{\Omega} 4 \left(\epsilon \mathcal{M}_{\alpha_1}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} - \epsilon \bar{\mathcal{M}}_{\alpha_1}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} \right)^2 d\Omega \\
& \leq C\epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[(\mathcal{M}_{\alpha_1}(\mathbf{y}) - \bar{\mathcal{M}}_{\alpha_1}(\mathbf{y})) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} \right]^2 d\Omega \\
& + C\epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\bar{\mathcal{M}}_{\alpha_1}(\mathbf{y}) \left(\frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} - \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} \right) \right]^2 d\Omega \quad (50) \\
& \leq C\epsilon^2 \epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} \right]^2 d\Omega + C\epsilon^2 \epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\bar{\mathcal{M}}_{\alpha_1}(\mathbf{y}) \right]^2 d\Omega \\
& \leq C\epsilon^2 \epsilon^2,
\end{aligned}$$

$$\begin{aligned}
& \sup_{0 \leq t \leq T^*} \int_{\Omega} 4 \left(\epsilon \mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} - \epsilon \bar{\mathcal{M}}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} \right)^2 d\Omega \\
& \leq C\epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[(\mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}) - \bar{\mathcal{M}}_{\alpha_1 \alpha_2}(\mathbf{y})) \frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} \right]^2 d\Omega \\
& + C\epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\bar{\mathcal{M}}_{\alpha_1 \alpha_2}(\mathbf{y}) \left(\frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} - \frac{\partial^2 \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} \right) \right]^2 d\Omega \\
& \leq C\epsilon^2 \epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} \right]^2 d\Omega + C\epsilon^2 \epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\bar{\mathcal{M}}_{\alpha_1 \alpha_2}(\mathbf{y}) \right]^2 d\Omega \\
& \leq C\epsilon^2 \epsilon^2, \quad (51)
\end{aligned}$$

$$\begin{aligned}
& \sup_{0 \leq t \leq T^*} \int_{\Omega} 4 \left(\epsilon \mathcal{R}_{\alpha_1}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} - \epsilon \bar{\mathcal{R}}_{\alpha_1}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial t} \right)^2 d\Omega \\
& \leq C\epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[(\mathcal{R}_{\alpha_1}(\mathbf{y}) - \bar{\mathcal{R}}_{\alpha_1}(\mathbf{y})) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} \right]^2 d\Omega \\
& + C\epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\bar{\mathcal{R}}_{\alpha_1}(\mathbf{y}) \left(\frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} - \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial t} \right) \right]^2 d\Omega \quad (52) \\
& \leq C\epsilon^2 \epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} \right]^2 d\Omega + C\epsilon^2 \epsilon^2 \sup_{0 \leq t \leq T^*} \int_{\Omega} \left[\bar{\mathcal{R}}_{\alpha_1}(\mathbf{y}) \right]^2 d\Omega \\
& \leq C\epsilon^2 \epsilon^2.
\end{aligned}$$

Next, substituting (49)-(52) into (48), we can get the final error estimation for inequality (48) as below.

$$\sup_{0 \leq t \leq T^*} \int_{\Omega} (T^{(2\epsilon)}(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t))^2 d\Omega \leq C\epsilon^2 + C\epsilon^2 \epsilon^2. \quad (53)$$

Analogously, we have the following inequality by virtue of the triangle inequality, mean value inequality, lemma 1 and assumption 1.

$$\begin{aligned}
& \int_0^{T^*} \|T^{(2\epsilon)}(\mathbf{x}, t) - \bar{T}^{(2\epsilon)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt \\
& \leq \int_0^{T^*} 4\|T^{(0)}(\mathbf{x}, t) - \bar{T}^{(0)}(\mathbf{x}, t)\|_{H^1(\Omega)}^2 dt \\
& + \int_0^{T^*} 4\|\epsilon \mathcal{M}_{\alpha_1}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}} - \epsilon \bar{\mathcal{M}}_{\alpha_1}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1}}\|_{H^1(\Omega)}^2 dt \\
& + \int_0^{T^*} 4\|\epsilon^2 \mathcal{M}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 T^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} - \epsilon^2 \bar{\mathcal{M}}_{\alpha_1 \alpha_2}(\mathbf{y}) \frac{\partial^2 \bar{T}^{(0)}(\mathbf{x}, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}}\|_{H^1(\Omega)}^2 dt \\
& + \int_0^{T^*} 4\|\epsilon^2 \mathcal{R}(\mathbf{y}) \frac{\partial T^{(0)}(\mathbf{x}, t)}{\partial t} - \epsilon^2 \bar{\mathcal{R}}(\mathbf{y}) \frac{\partial \bar{T}^{(0)}(\mathbf{x}, t)}{\partial t}\|_{H^1(\Omega)}^2 dt \\
& \leq C\epsilon^2 + C\epsilon^2\epsilon^2.
\end{aligned} \tag{54}$$

Finally, we incorporate the above inequalities (53) and (54) into inequality (47) to complete the error estimate (46), which ensures the numerical convergence of the proposed HOMS-RNN method.

5. Numerical experiments and discussions

This section conducts a suite of numerical experiments, encompassing high-contrast composites, three-dimensional composites, porous materials as well as large-scale composites, to validate the effectiveness and robustness of the presented HOMS-RNN method. All numerical examples are implemented via single-hidden layer RNNs with the Tanh activation function, whose initialized weights and variance are set in the range $[-R_m, R_m] = [-5, 5]$. Moreover, the hard-constrained boundary method is employed to impose the Dirichlet boundary conditions for all RNNs approximating the solutions of matrix material. Furthermore, the uniform sampling approach is used to generate the residual points within and at the boundaries of material region. In order to further improve the efficiency of the proposed HOMS-RNN approach, the following first-order and second-order central difference schemes are utilized to approximate the first-order and second-order partial derivatives involving in multi-scale simulation respectively, please refer to [34, 35].

$$\begin{aligned}
\frac{\partial T(x_1, x_2, t)}{\partial x_1} &= \frac{T(x_1 + h, x_2, t) - T(x_1 - h, x_2, t)}{2h}, \\
\frac{\partial^2 T(x_1, x_2, t)}{\partial^2 x_1} &= \frac{T(x_1 + h, x_2, t) - 2T(x_1, x_2, t) + T(x_1 - h, x_2, t)}{h^2},
\end{aligned}$$

where the step sizes for the first-order and second-order central differences are defined as $h = 10^{-6}$ and $h = 5 \times 10^{-4}$, respectively. Next, to more conveniently evaluate the computational accuracy of the proposed HOMS-RNN method, we

introduce the following error notations. Among these error notations, T^ϵ represents the reference numerical solution for multi-scale problem obtained by direct finite element simulation (linear conforming finite element method) on fine grid and finite difference simulation (first-order backward difference scheme) in time-domain with time step 0.002. $\bar{T}^\epsilon$ denotes the reference neural network solution obtained by direct RNN simulation.

$$\begin{aligned} Lerror_1 &= \frac{\|T^{(0)} - T^\epsilon\|_{L^2}}{\|T^\epsilon\|_{L^2}}, Lerror_2 = \frac{\|T^{(1\epsilon)} - T^\epsilon\|_{L^2}}{\|T^\epsilon\|_{L^2}}, Lerror_3 = \frac{\|T^{(2\epsilon)} - T^\epsilon\|_{L^2}}{\|T^\epsilon\|_{L^2}}, \\ Lerror_4 &= \frac{\|\bar{T}^{(2\epsilon)} - T^\epsilon\|_{L^2}}{\|T^\epsilon\|_{L^2}}, Lerror_5 = \frac{\|\bar{T}^{(2\epsilon)} - T^{(2\epsilon)}\|_{L^2}}{\|T^{(2\epsilon)}\|_{L^2}}, Lerror_6 = \frac{\|\bar{T}^\epsilon - T^\epsilon\|_{L^2}}{\|T^\epsilon\|_{L^2}}, \\ Herror_1 &= \frac{\|T^{(0)} - T^\epsilon\|_{H^1}}{\|T^\epsilon\|_{H^1}}, Herror_2 = \frac{\|T^{(1\epsilon)} - T^\epsilon\|_{H^1}}{\|T^\epsilon\|_{H^1}}, Herror_3 = \frac{\|T^{(2\epsilon)} - T^\epsilon\|_{H^1}}{\|T^\epsilon\|_{H^1}}, \\ Herror_4 &= \frac{\|\bar{T}^{(2\epsilon)} - T^\epsilon\|_{H^1}}{\|T^\epsilon\|_{H^1}}, Herror_5 = \frac{\|\bar{T}^{(2\epsilon)} - T^{(2\epsilon)}\|_{H^1}}{\|T^{(2\epsilon)}\|_{H^1}}, Herror_6 = \frac{\|\bar{T}^\epsilon - T^\epsilon\|_{H^1}}{\|T^\epsilon\|_{H^1}}. \end{aligned}$$

In addition, all numerical experiments are implemented on a desktop workstation equipped with an Intel(R) Xeon(R) Gold 6146 CPU of 3.20 GHz, an internal memory of 96.0 GB and a NVIDIA GeForce RTX 3080 GPU card.

5.1. Example 1: the drawback analysis of direct RNN simulation for multi-scale problems

The neural network-based methods have been employed to address a multitude of equations. Nonetheless, their efficacy in tackling multi-scale problems has been found to be compromised. To illustrate this, direct RNN simulation is implemented for heat conduction equation (1) of composite materials with different parameter ϵ . If there are no special instructions, all numerical experiments in this study set internal heat source $f(\mathbf{x}, t) = 10$, boundary condition $\hat{T}(\mathbf{x}, t) = 0$, initial condition $\check{T}(\mathbf{x}) = 0$ and $T^* = 5.0$. The microstructure of the investigated composites is exhibited in Fig. 1. The material property parameters of matrix and inclusion constituents are given in Table 1.

Table 1: Material property parameters.

Property	Matrix	Inclusion
k_{ij}^ϵ	10.0	1.0
ρ^ϵ	2.0	1.0
c^ϵ	1.0	1.0

In this numerical experiment, the matrix region of composite materials is simulated using RNNs that possess a hidden layer consisting of 500 neurons, while each inclusion region is separately simulated with RNNs featuring a hidden layer of 200 neurons. Besides, the setup entailed sampling 3,000 residual points within each microscopic unit cell and 1,000 residual points along the interface

between the matrix and inclusion components. Moreover, the adjustable weight parameter in the linear system of RNNs is defined as 100.

Limited by the substantial computational resources for directly RNN simulating multi-scale problems, we only simulate multi-scale problem (1) with small parameter $\epsilon = 1, 1/2, 1/3, 1/4, 1/5$ cases. Fig. 5 displays the computational results at time $t = 5.0$ of reference solutions from fine FEM computations, as well as direct RNN simulations. It is evident from Fig. 5 that as the scale of composite materials increases, direct RNN simulations are not effectively capable of simulating the transient multi-scale heat conduction problem, especially for accurately capturing the highly oscillatory information at micro-scale. Furthermore, Table 2 presents the computational time for direct RNN simulations as ϵ varies, indicating that as ϵ decreases, the simulative time required for direct RNN approach increases dramatically. Fig. 6 depicts the evolutive relative errors of the temperature field of composite materials with periodic parameter $\epsilon = 1/5$, which clearly indicates that the computational accuracy of the direct RNN approach is extremely bad and unacceptable for practical applications.

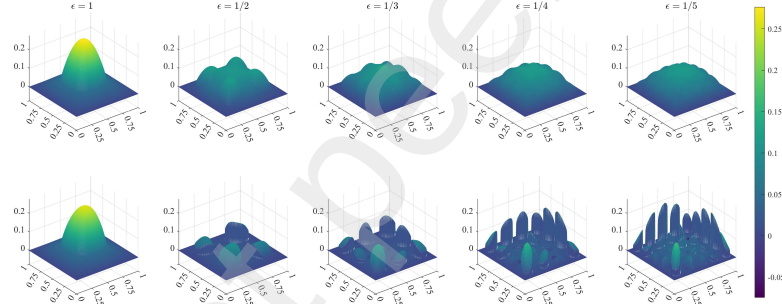


Figure 5: The computing results of temperature field inside composite materials. The first row displays the numerical results of fine FEM computation, and the second row displays the numerical results of direct RNN computation.

Table 2: The computational time of direct RNN simulations.

ϵ	1	1/2	1/3	1/4	1/5
time	2.33s	11.89s	55.26s	572.61s	1830.26s

In conclusion, the raw RNN method is ineffective for multi-scale problem (1) of composite materials due to prohibitive computation and Frequency Principle. Hence, developing effective neural network-based approach is of great application value and theoretical significance.

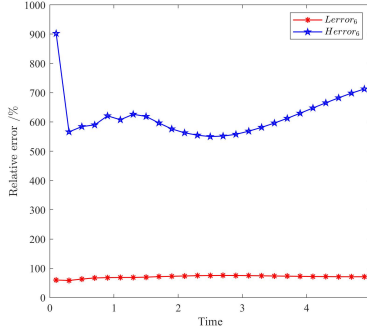


Figure 6: The evolving relative errors of temperature field inside composite materials.

5.2. Example 2: transient multi-scale heat simulation of composite materials with different thermal conductivities

This example utilizes the HOMS-RNN approach to simulate the transient multi-scale heat conduction problem of composite materials with distinct thermal conductivity coefficients. The microstructure of the investigated materials is illustrated in Fig. 1, where the characteristic size $\epsilon = 1/5$. The material parameters ρ^ϵ and c^ϵ for both matrix and inclusion materials are detailed in Table 1. The thermal conductivity coefficient k_{ij}^ϵ of inclusion material is defined as 1.0, and the thermal conductivity coefficient k_{ij}^ϵ of matrix material is defined as 10.0, 50.0 and 100.0 respectively.

In this example, before implementing transient multi-scale heat simulation of composite materials, the computational effects of hard-constraint and soft-constraint boundary conditions for the proposed HOMS-RNN approach are compared. Firstly, the smoothing parameter of material transition k in formula (32) is set as 300. To approximate the microscopic cell functions $\mathcal{M}_{\alpha_1}$, $\mathcal{M}_{\alpha_1\alpha_2}$ and $\mathcal{R}$, we employ RNNs with 320 neurons in the hidden layer for matrix and inclusion regions of composite materials. Moreover, we sample 9,000 residual points within microscopic cell region and 3,000 residual points along the interface between matrix and inclusion materials. Furthermore, the adjustable weight parameters λ_2 and λ_3 in the linear system of microscopic cell functions with hard-constraint boundary condition are set as 1,000. To compute the macroscopic homogenized coefficients, a total of 20,000 points are sampled within microscopic unit cell Y . Next, we assign 150,000 residual points within macroscopic homogenized spatial-temporal domain $\Omega \times (0, T^*)$. And also, RNN with 640 neurons in the hidden layer is utilized to approximate the macroscopic homogenized solutions of multi-scale heat equation (1). Comparatively, for the soft-constraint boundary condition, 9,000 residual points are assigned on the boundary ∂Y , and the adjustable parameters λ_1 , λ_2 and λ_3 in the linear system are set to 1,000 when calculating microscopic cell functions. Moreover, when computing macroscopic homogenized solution, we sample 50,000 residual points on the boundary

$\partial\Omega$, 25,000 residual points on the initial domain Ω , and the adjustable weight parameters of γ_1 and γ_2 in the linear system of macroscopic homogenized solution are defined as 1,000. After that, Table 3 illustrates the accuracy results of hard-constraint and soft-constraint boundary conditions for microscopic cell functions and the macroscopic homogenized solution at time $t = 5.0$, which clearly demonstrate the computing performance of hard-constraint boundary condition is better than soft-constraint boundary condition.

Table 3: The comparison of hard-constraint and soft-constraint boundary conditions.

	$\mathcal{M}_1$	$\mathcal{M}_{11}$	$\mathcal{M}_{12}$	$\mathcal{R}$	$T^{(0)}$
Hard-constraint	2.6%	0.72%	2.4%	0.53%	1.12%
Soft-constraint	2.9%	2.0%	3.5%	0.54%	5.91%

After multi-scale simulation of composite materials, Fig. 7 depicts the computational results of HOMS-RNN simulation and reference FEM simulation for microscopic cell functions $\mathcal{M}_{\alpha_1}$, $\mathcal{M}_{\alpha_1\alpha_2}$, $\mathcal{R}$ and macroscopic homogenized solution $T^{(0)}$ at time $t = 5.0$. From the figure, it can be clearly seen that the proposed HOMS-RNN approach can accurately solve both microscopic cell functions and macroscopic homogenized solution. And then, the automatic differentiation technique of neural networks is employed to assemble microscopic cell functions and macroscopic homogenized solution into the HOMS-RNN solution. Fig. 8 illustrates the macroscopic homogenized solution, lower-order multi-scale solution, higher-order multi-scale solution, reference FEM solution and HOMS-RNN solution for multi-scale heat problem (1) of composite material with thermal conductivity contrast ratio 10 at time $t = 5.0$. In addition, Fig. 9 visualizes the evolutive relative errors of the temperature field by different approaches. According to Figs. 8 and 9, it is apparent that the proposed HOMS-RNN method effectively captures the microscopic oscillatory phenomena of composite material, and exhibits consistent stability throughout the whole temporal evolution.

Furthermore, considering the high-contrast cases of composite materials, two groups of composite materials with thermal conductivity contrast ratio 50 and 100 are simulated by the proposed HOMS-RNN method. The neural network parameters of the HOMS-RNN approach for contrast ratio 50 and 100 are the same as the experimental settings of contrast ratio 10 described above. Figs. 10 and 11 display the macroscopic homogenized solution, lower-order multi-scale solution, higher-order multi-scale solution, reference FEM solution and HOMS-RNN solution at time $t = 5.0$ for thermal conductivity contrast ratio 50 and 100, respectively. Figs. 12 and 13 further display the evolutionary relative errors of the temperature field for contrast ratio 50 and 100. As illustrated in these figures, it can be obviously seen that as the contrast ratio of material coefficients increases, the lower-order multi-scale method cannot effectively capture the microscopic oscillatory phenomenon of heterogeneous materials. Conversely, by resolving the prohibitive computation and Frequency Principle, the HOMS-RNN method

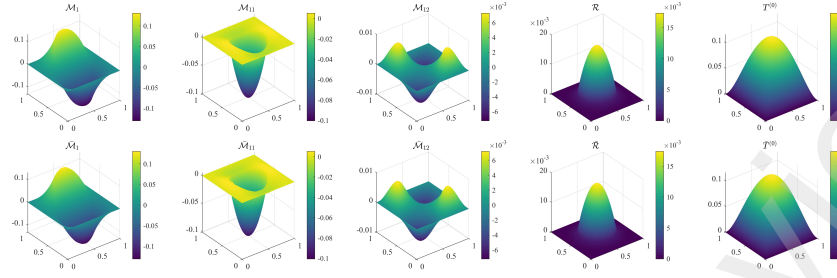


Figure 7: The numerical results of microscopic cell problems and macroscopic homogenized problem with contrast ratio 10. The first and second rows exhibit the numerical results obtained by FEM and HOMS-RNN, respectively.

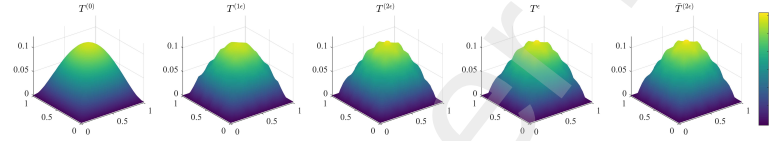


Figure 8: The computational results of temperature field with contrast ratio 10 at $t = 5.0$.

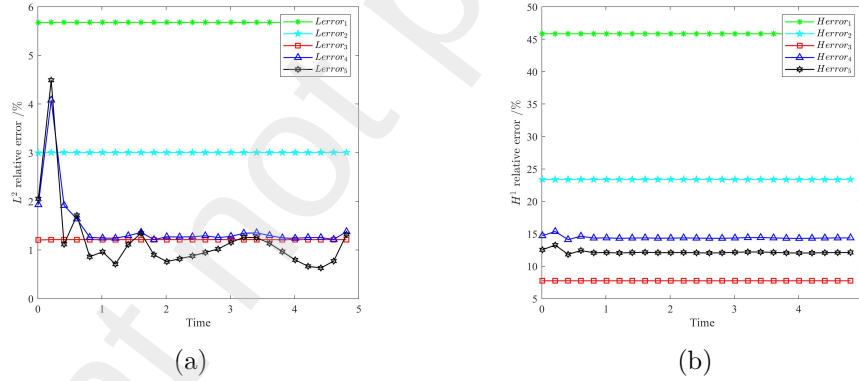


Figure 9: The evolving relative errors of temperature field with contrast ratio 10. (a) L^2 norm error $Lerror$. (b) H^1 norm error $Herror$.

can mesh-free and accurately capture the highly microscopic oscillations, and showcases stable performance over the temporal domain. Moreover, numerical results indicate the numerical accuracy of the HOMS-RNN approach is comparable and competitive to that of FEM, especially for high-contrast composite materials.

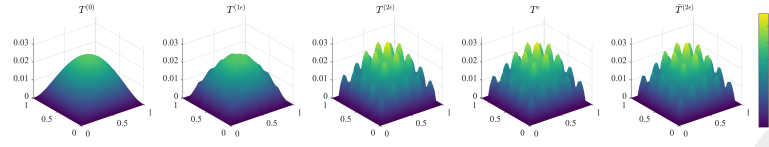


Figure 10: The computing results of temperature field with thermal conductivity contrast ratio 50.

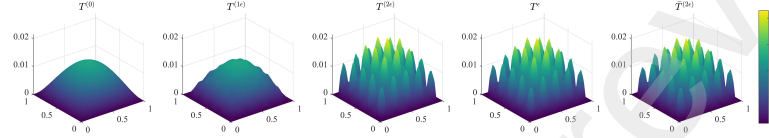


Figure 11: The computing results of temperature field with thermal conductivity contrast ratio 100.

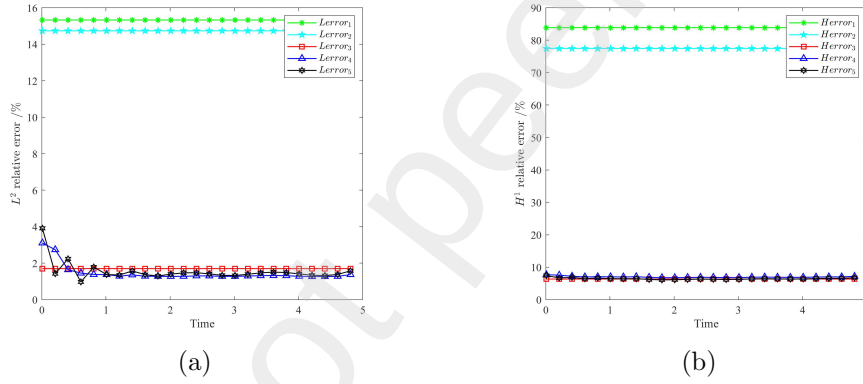


Figure 12: The evolving relative errors of temperature field with thermal conductivity contrast ratio 50. (a) L^2 norm error $Lerror$. (b) H^1 norm error $Herror$.

5.3. Example 3: the robustness analysis of HOMS-RNN method

This example examines the robustness of the proposed HOMS-RNN approach by progressively diminishing the characteristic parameter ϵ of composite materials, resulting in $O((1/\epsilon)^2)$ microscopic unit cells in 2D composite materials. Employing the composite materials with thermal conductivity contrast ratio 50 in subsection 5.2, we continue to simulate their multi-scale thermal behaviors when setting different characteristic parameter ϵ .

According to HOMS-RNN solution (42) of HOMS-RNN approach, the microscopic cell functions and macroscopic homogenized solution obtained in subsection 5.2 are directly utilized for the assembly of the HOMS-RNN solution with different characteristic parameter ϵ . Fig. 14 illustrates the numerical re-

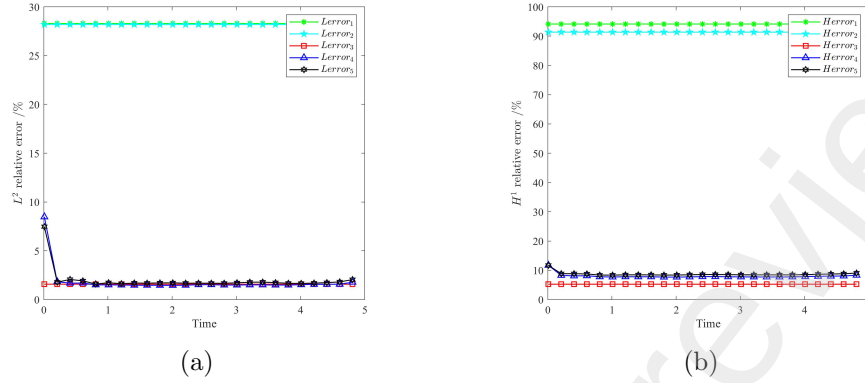


Figure 13: The evolving relative errors of temperature field with thermal conductivity contrast ratio 100. (a) L^2 norm error $Lerror$. (b) H^1 norm error $Herror$.

sults of the homogenized method, lower-order multi-scale method, higher-order multi-scale method, finite element method and HOMS-RNN method for different characteristic parameters at the moment $t = 5.0$. It can be observed from this figure that the HOMS-RNN method can accurately capture the highly oscillatory phenomena of composite materials at micro-scale even the number of microscopic unit cells in 2D composite materials dramatically improves, while the oscillation amplitudes consistent with those obtained from higher-order multi-scale approach and reference FEM. Comparatively, direct deep learning simulation for large-scale composite materials with massive microscopic unit cells will expend a tremendous amount of computational resources, and even no convergence due to the Frequency Principle. Fig. 15 further visualizes the evolutive relative errors of $Lerror_4$ and $Herror_4$ under different characteristic parameters. The corresponding results manifest that the numerical accuracy of the HOMS-RNN approach is competitive with reference FEM although the characteristic parameter progressively diminishes, and also validate the robust stability of the HOMS-RNN approach proposed in this study. Moreover, Table 4 lists the computational time of the higher-order multi-scale method, finite element method and HOMS-RNN method. The time comparison in Table 4 indicates that the computational efficiency of the proposed HOMS-RNN method is significantly better than that of higher-order multi-scale approach and reference FEM on fine grid. Particularly, as periodic parameter ϵ decreases, the efficiency of finite element method increases dramatically, whereas the computational time for the HOMS-RNN method remains relatively stable. This prominent advantage of low computational overhead of HOMS-RNN method enables the effective simulation and analysis of multi-scale thermal transfer behaviors of large-scale composite materials.

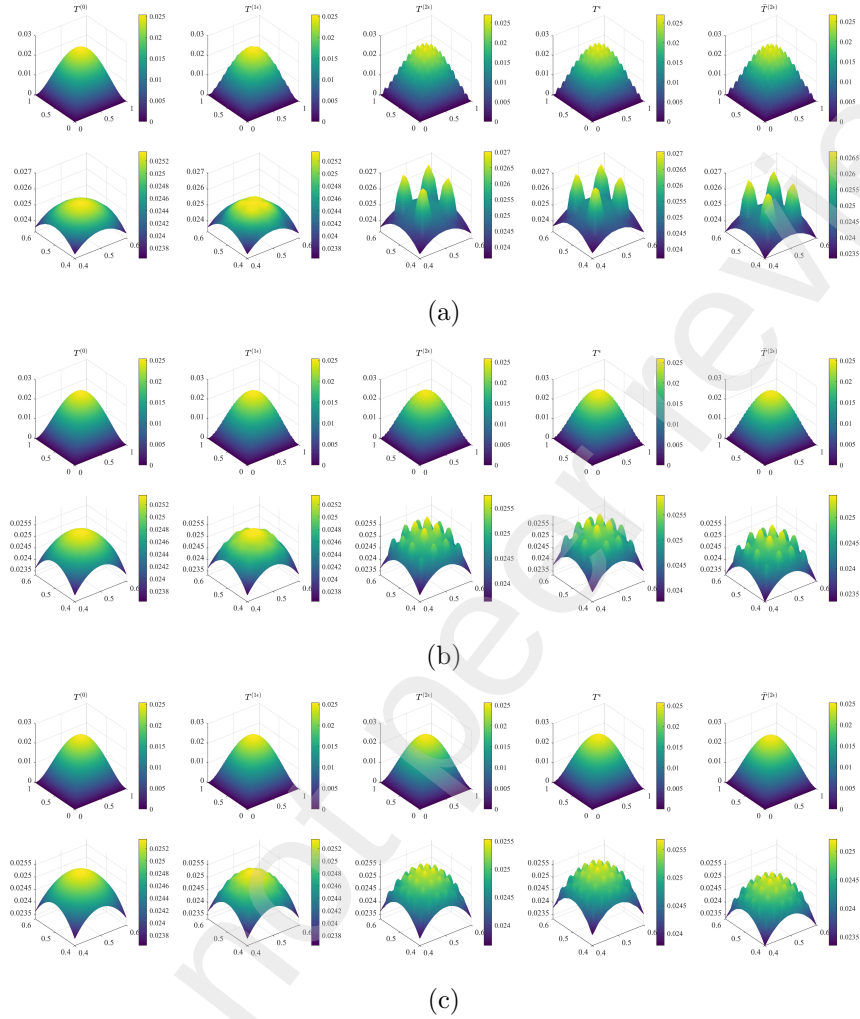


Figure 14: The evolving relative errors of temperature field for different characteristic parameters. (a) $\epsilon = 1/10$. (b) $\epsilon = 1/20$. (c) $\epsilon = 1/30$.

Table 4: The computational time for different characteristic parameter ϵ .

ϵ	1/5	1/10	1/20	1/30
HOMS-RNN method	96.86s	97.23s	100.75s	106.46s
HOMS method	134.80s	160.58s	286.73s	440.64s
FEM	144.81s	591.12s	2611.13s	7016.75s

5.4. Example 4: applications of HOMS-RNN method for transient heat conduction simulation of 3D composite materials

In this example, we apply the presented HOMS-RNN approach to simulate 3D transient multi-scale heat conduction problem. The geometric configura-

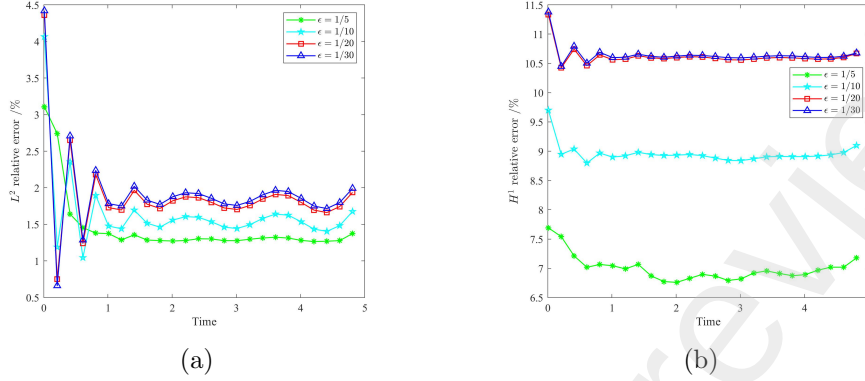


Figure 15: The evolving relative errors of temperature field for different characteristic parameters. (a) L^2 norm error $Lerror$. (b) H^1 norm error $Herror$.

tion of the investigated 3D heterogeneous material is shown in Fig. 16, and macroscopic structure is dominated by $\Omega = (x_1, x_2, x_3) = [0, 1] \times [0, 1] \times [0, 1]$ cm^3 with characteristic size $\epsilon = 1/5$. Meanwhile, this example considers actual SiC/C composite materials, whose material parameters are presented in Table 5. The function that smoothly approximates the treatment of discontinuous material parameters is specified as follows.

$$f_k = (c_1 - c_2) f_s \left(k \left((y_1 - o_1)^2 + (y_2 - o_2)^2 + (y_3 - o_3)^2 - r^2 \right) \right) + c_2, \quad (55)$$

where (o_1, o_2, o_3) denotes the central coordinate of spherical inclusion, which is assigned as $(0.5, 0.5, 0.5)cm$. The variable r denotes the radius of spherical inclusion, which is assigned as $0.3cm$. Next, the material transition smoothing parameter k is defined as 200. Moreover, the internal heat source of 3D composite materials is defined by $f(\mathbf{x}, t) = 10J/(cm^3 \cdot s)$, boundary condition $\hat{T}(\mathbf{x}, t) = 0$, initial condition $\dot{T}(\mathbf{x}) = 0$ and total simulation time $T^* = 5.0s$.

Table 5: The material parameters of heterogeneous SiC/C composite.

Property	Matrix SiC	Inclusion C
$k_{ij}^\epsilon (W/(m \cdot K))$	258.18	15.61
$\rho^\epsilon (kg/m^3)$	3210	1760
$c^\epsilon (J/(kg \cdot K))$	1100.31	1066.46

For the purpose of simulating microscopic cell functions $\mathcal{M}_{\alpha_1}$, $\mathcal{M}_{\alpha_1\alpha_2}$ and $\mathcal{R}$, RNNs with 1,280 neurons in the hidden layer are employed to individually approximate microscopic cell functions in matrix and inclusion regions of composite materials. Besides, a total of 40,000 residual points are sampled within microscopic unit cell, while an additional 20,000 residual points are distributed along the interface boundary separating matrix and inclusion materials. The

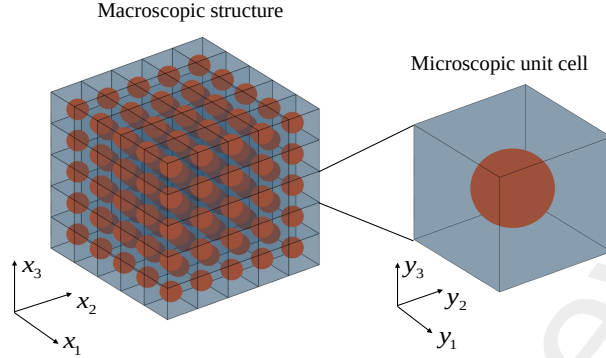


Figure 16: The schematic of 3D composite material.

adjustable weight parameter λ_2 and λ_3 within the linear system of microscopic cell functions are assigned a value of 1,000. After computing lower-order microscopic cell functions, an aggregate of 200,000 points in the realm of lower-order microscopic cell functions is sampled for the computation of the macroscopic homogenized coefficients. Furthermore, in order to solve macroscopic homogenized solution of 3D composite materials, the number of residual points is set to 250,000 for spatial-temporal domain $\Omega \times (0, T^*)$. Then, an RNN with 1,280 neurons in the hidden layer is utilized to solve macroscopic homogenized equation of 3D composite materials. After solving all microscopic cell functions and macroscopic homogenized solution, the automatic differentiation technique of neural networks is employed to assemble the HOMS-RNN solution for 3D composite materials. Fig. 17 depicts the macroscopic homogenized solution, lower-order multi-scale solution, higher-order multi-scale solution, reference FEM solution and the HOMS-RNN solution at the time point $t = 5.0$. Fig. 18 further displays the evolutive relative errors of temperature field in 3D composite material.

As exhibited in Fig. 17, it can be clearly found that the proposed HOMS-RNN approach can accurately capture microscopic high-frequency oscillatory response of 3D composite material due to its spatial heterogeneity, which is in excellent agreement with the reference FE solution. However, the classical homogenized method can only catch the macroscopic material behavior without oscillations, lower-order multi-scale method can capture inadequately low-frequency material response at micro-scale. In addition, the numerical results in Fig. 18 plainly demonstrate the proposed HOMS-RNN approach for 3D time-dependent multi-scale simulation is stable and robust after long-time numerical simulation. Additionally, it should be highlighted that the 3D multi-scale simulation using higher-order multi-scale method requires $7.1h$, the 3D simulation by reference FEM expends $11.8h$, whereas the HOMS-RNN method developed in this study only necessitates $1.5h$, which greatly improves the computational efficiency and is cost-effective for practical engineering application.

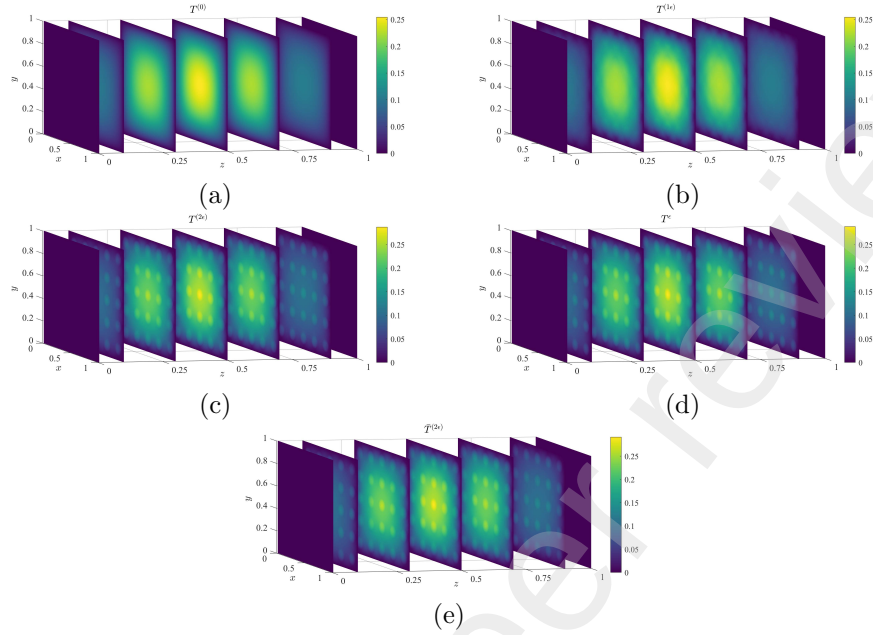


Figure 17: The computational results of temperature field in 3D composite material. (a) The macroscopic homogenized solution $T^{(0)}$. (b) The LOMS solution $T^{(1\epsilon)}$. (c) The HOMS solution $T^{(2\epsilon)}$. (d) The FEM solution T^ϵ . (e) The HOMS-RNN solution $\bar{T}^{(2\epsilon)}$.

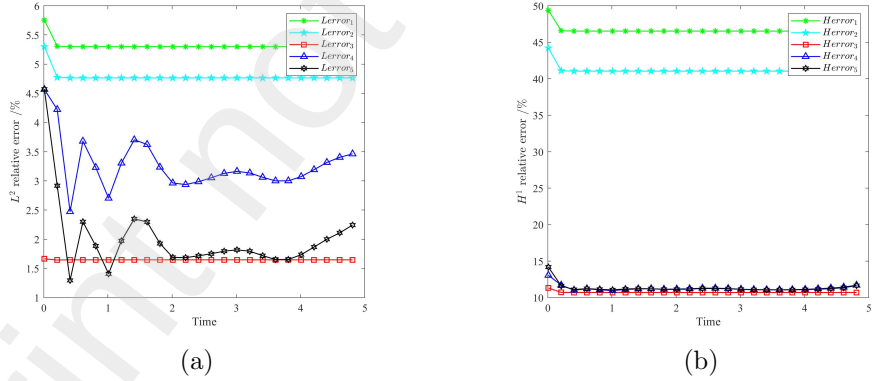


Figure 18: The evolving relative errors of temperature field in 3D composite material. (a) L^2 norm error $Lerror$. (b) H^1 norm error $Herror$.

5.5. Example 5: applications of HOMS-RNN for transient heat conduction simulation of porous materials

This example validates the effectiveness of the proposed HOMS-RNN approach for porous materials. Considering the transient multi-scale heat conduc-

tion problem (1) for porous materials, the following boundary condition should be imposed on the internal pore boundary Γ_q^ϵ of porous materials.

$$n_i k_{ij}^\epsilon(\mathbf{x}) \frac{\partial T^\epsilon(\mathbf{x}, t)}{\partial x_j} = 0, \text{ on } \Gamma_q^\epsilon \times (0, T^*),$$

where n_i represents the normal vector on internal pore boundary Γ_q^ϵ .

Through the higher-order multi-scale modeling, we can derive the higher-order multi-scale asymptotic solution for transient multi-scale heat conduction problem (1) of porous materials, whose formula is the same as formula (20). The only difference is that the following boundary condition is imposed on internal pore boundary Γ_q of microscopic unit cell for lower-order cell $\mathcal{M}_{\alpha_1}(\mathbf{y})$.

$$n_i k_{ij}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_1}(\mathbf{y})}{\partial y_j} = -n_i k_{i\alpha_1}(\mathbf{y}), \text{ on } \Gamma_q.$$

For higher-order cell $\mathcal{M}_{\alpha_1\alpha_2}(\mathbf{y})$, the following boundary condition is added on internal pore boundary Γ_q of microscopic unit cell.

$$n_i k_{ij}(\mathbf{y}) \frac{\partial \mathcal{M}_{\alpha_1\alpha_2}(\mathbf{y})}{\partial y_j} = -n_i k_{i\alpha_2}(\mathbf{y}) \mathcal{M}_{\alpha_1}(\mathbf{y}), \text{ on } \Gamma_q.$$

And also, for higher-order cell $\mathcal{R}(\mathbf{y})$, the following boundary condition is added on internal pore boundary Γ_q of microscopic unit cell.

$$n_i k_{ij}(\mathbf{y}) \frac{\partial \mathcal{R}(\mathbf{y})}{\partial y_j} = 0, \text{ on } \Gamma_q.$$

The porous materials consist of only one type of material, whose material property parameters are defined as follows: thermal conductivity coefficient $k_{ij}^\epsilon(\mathbf{x}) = 1$, mass density $\rho^\epsilon(\mathbf{x}) = 1$ and specific heat $c^\epsilon(\mathbf{x}) = 1$. Firstly, we investigate porous material I with a characteristic size $\epsilon = 1/5$, as illustrated in Fig. 19. Next, RNNs with 640 neurons in the hidden layer are utilized for solving microscopic cell functions. The RNN computation involves sampling 7,000 residual points within microscopic cell region and 3,000 residual points along the internal pore boundary of microscopic unit cell. The adjustable weight parameter λ_1 in the linear system of microscopic cell functions is set to 10. After that, a total of 10,000 points within microscopic unit cell are selected to calculate the macroscopic homogenized coefficients. Moreover, assigning 150,000 residual points in the spatial-temporal region $\Omega \times (0, T^*)$, an RNN with 640 neurons in the hidden layer is used to solve macroscopic homogenized equation of porous material I. The automatic differentiation technique of neural networks is further employed to compute the HOMS-RNN solution of porous material I. Fig. 20 displays the computational results at $t = 5.0$ of both classical numerical methods and HOMS-RNN method. Fig. 21 illustrates the evolutive relative errors of different methods. It can be observed from the figures that for porous material I, the numerical accuracy of the HOMS-RNN approach proposed in this

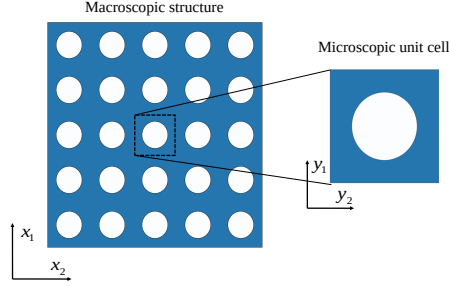


Figure 19: The schematic of porous material I.

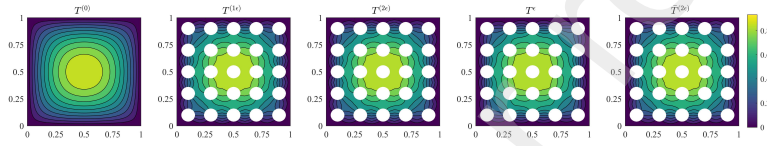


Figure 20: The computational results of temperature field of porous material I at $t = 5.0$.

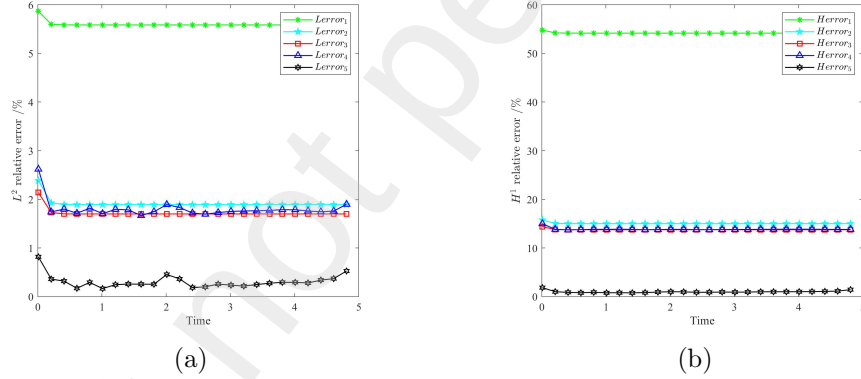


Figure 21: The evolving relative errors of temperature field of porous material I. (a) L^2 norm error $Lerror$. (b) H^1 norm error $Herror$.

work is nearly equivalent to that of classical FEM and higher-order multi-scale method.

Next, we verify the robustness of the HOMS-RNN method for porous material I by setting different characteristic size $\epsilon = 1/10, 1/20$ and $1/30$. Fig. 22 displays the evolving relative errors $Lerror_4$ and $Herror_4$ with respect to characteristic size ϵ . It is evident from the figure that as ϵ decreases, the computational errors of the HOMS-RNN method exhibit a trend of convergence. Furthermore, Table 6 lists the computational time for transient thermal sim-

ulation of porous material I. It can be drawn from the Table 6 that the time consumption of novel HOMS-RNN method is much lower than that of HOMS method and classical FEM. Moreover, as ϵ decreases, the computational efficiency of classical FEM and HOMS methods is gradually reduced, while the computational efficiency of novel HOMS-RNN method is almost unchanged.

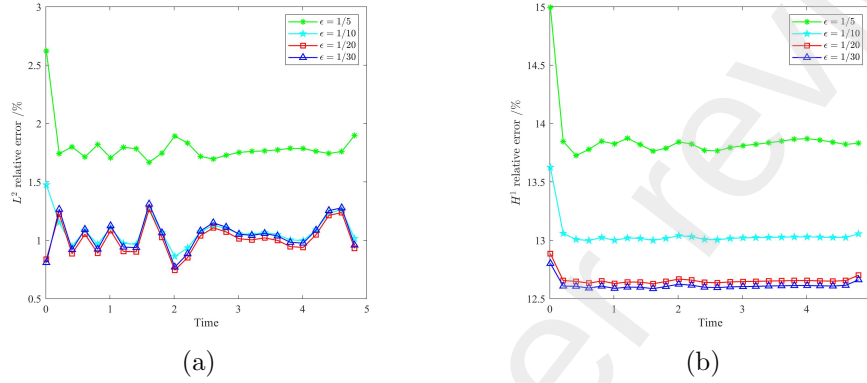


Figure 22: The evolving relative errors of temperature field of porous material I with different characteristic parameters. (a) L^2 norm error $Error$. (b) H^1 norm error $Error$.

Table 6: The computational time of porous material I with different characteristic parameter ϵ .

ϵ	1/5	1/10	1/20	1/30
HOMS-RNN method	72.88s	74.83s	75.84s	79.46s
HOMS method	165.22s	204.80s	354.05s	614.68s
FEM	253.54s	1182.69s	6603.60s	23758.15s

In order to further verify the application value of the proposed HOMS-RNN approach, the time-dependent thermal response of complex porous material II is investigated, whose geometric configuration is depicted in Fig. 23. Next, RNNs with a hidden layer of 2,000 neurons are employed for solving microscopic cell functions. Then, we assign 5,000 residual points within microscopic cell region and 7,000 residual points along the internal pore boundary of microscopic unit cell, and the adjustable weight parameter λ_1 in the linear system of microscopic cell functions is defined as 10. Afterwards, 10,000 points are selected within microscopic unit cell to calculate macroscopic homogenized coefficients. Setting 150,000 residual points in homogenized spatial-temporal region $\Omega \times (0, T^*)$, an RNN with a hidden layer containing 640 neurons are further employed to compute macroscopic homogenized equation. Finally, the automatic differentiation approach of neural networks is applied to integrate the microscopic cell functions and macroscopic homogenized solution into the HOMS-RNN solution for porous

material II. Fig. 20 unveils the computational results of traditional numerical methods and HOMS-RNN method at $t = 5.0$. Fig. 21 illustrates the evolutive relative errors of different computational methods. It can be clearly observed from the figures that for complex porous material II, the numerical accuracy of the HOMS-RNN method presented in this study is competitive with traditional numerical methods. And also, the microscopic oscillating information can be accurately captured by the proposed HOMS-RNN method.

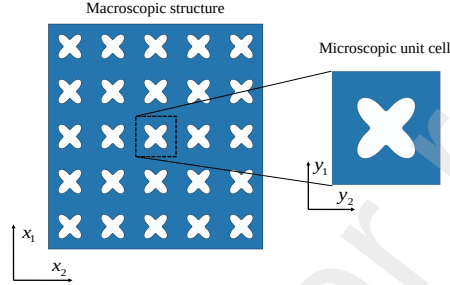


Figure 23: The schematic of porous material II.

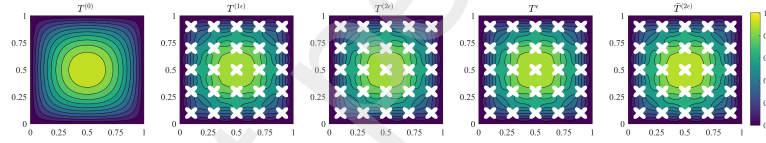


Figure 24: The computing results of temperature field of porous material II at $t = 5.0$.

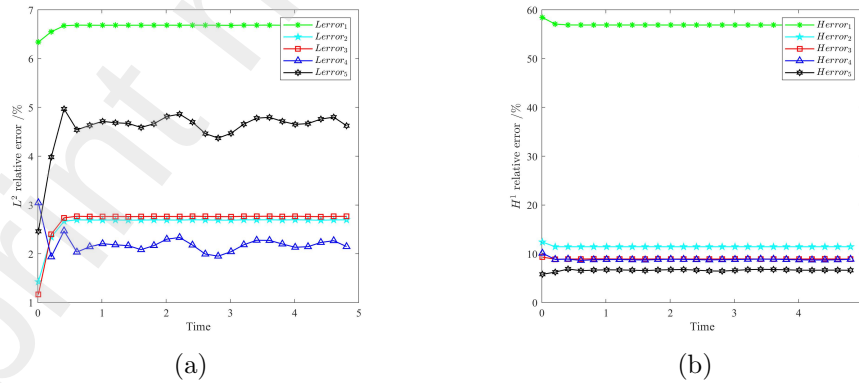


Figure 25: The evolving relative errors of temperature field of porous material II. (a) L^2 norm error $Lerror$. (b) H^1 norm error $Herror$.

Subsequently, the robustness of the presented HOMS-RNN approach for complex porous material II is validated by assigning characteristic size ϵ as 1/10, 1/20 and 1/30. Fig. 26 illustrates the evolution relative errors of L^2 and H^1 with respect to the characteristic size ϵ . It can be obviously found that as ϵ diminishes, the computational errors of HOMS-RNN approach tend to converge. The time consumption for HOMS-RNN method, as presented in Table 7, is significantly less than that of HOMS method and classical FEM. Furthermore, with the increase of material scale, the computational efficiency of classical FEM and HOMS method is gradually reduced, whereas the computational efficiency of HOMS-RNN method remains relatively constant. It is of great practical value for large-scale engineering thermal simulation.

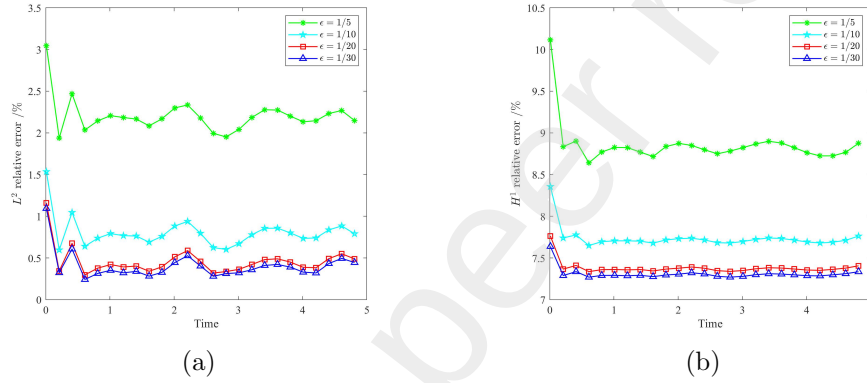


Figure 26: The evolving relative errors of temperature field of porous material II with different characteristic parameters. (a) L^2 norm error L^2 . (b) H^1 norm error H^1 .

Table 7: The computational time of porous material II with different characteristic parameter ϵ .

ϵ	1/5	1/10	1/20	1/30
HOMS-RNN method	225.81s	231.01s	233.86s	291.85s
HOMS method	292.61s	351.44s	632.27s	1010.36s
FEM	193.97s	1031.76s	5722.09s	17657.60s

6. Conclusions and outlooks

In this study, an innovative HOMS-RNN methodology is developed for efficiently solving complex multi-scale parabolic problems of authentic heterogeneous materials with highly discontinuous and high-contrast material parameters, that inherits the high-accuracy and model reduction advantages of the

higher-order multi-scale method, and the mesh-free, high-efficiency and space-time advantages of randomized neural networks. With the help of higher-order multi-scale approach, original physical constraints (multi-scale PDEs) of neural networks is decoupled as new physical constraints (macroscopic homogenized PDEs and microscopic cell PDEs) of neural networks, which tackle the exorbitant computation and Frequency Principle of multi-scale simulation. Moreover, high-order microscopic cell functions and corresponding correction terms substantially improve the numerical accuracy of multi-scale heat simulation and can precisely capture the highly microscopic oscillations due to spatial heterogeneities of inhomogeneous materials. Furthermore, randomized neural networks are employed and improved as numerical solvers, which can mesh-free and efficiently solve microscopic cell problems, and space-time solve macroscopic homogenized problem with high-accuracy, high-efficiency and mesh-free advantages. Additionally, the convergence result of the proposed HOMS-RNN approach is rigorously demonstrated under some appropriate assumptions, that can theoretically ensure the computational convergence of the proposed approach.

The numerical results of numerical experiments clearly testify that the proposed HOMS-RNN approach offer several advantages: (i) achieving high-accuracy and mesh-free performance for multi-scale heat simulation compared with FEM and multi-scale method, especially for accurately capturing the microscopic oscillatory behaviors of composite materials; (ii) possessing efficient and space-time capability for time-dependent multi-scale problems, which is competitive with traditional numerical methods; (iii) maintaining the robustness and the numerical accuracy for large-scale heat simulation without increasing computational complexity. These computational strengths are of significant significance and potential for practical engineering applications.

In the future, several aspects of these methods are required further investigation. First immediate continuation of this HOMS-RNN framework is extended to multi-physics coupling problems of composite materials. It is non-trivial and currently under investigation, and it will be addressed in a future publication. Another worthy continuation is that leveraging parallel superiority of randomized neural network further improve the computational efficiency. Furthermore, this framework will be extended to nonlinear and random multi-scale problems. These problems are difficult to handle and new numerical techniques should be introduced into the proposed HOMS-RNN framework.

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Data availability

Inquiries about data availability should be directed to the authors.

Declarations

The authors affirm that they have no known financial or interpersonal conflicts that might have looked to have influenced the research presented in this study.

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